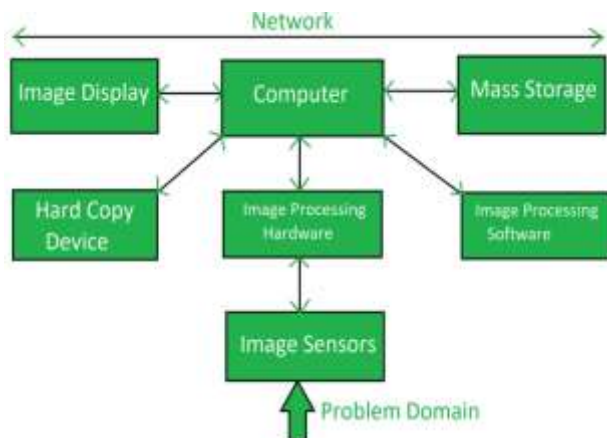
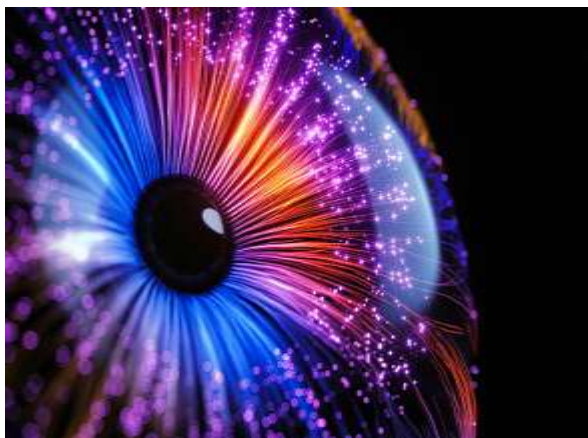


# Digital Image Processing



**Dr. Md. Aynal Haque**

**Professor, EEE, BUET**

**Email:** [aynal@eee.buet.ac.bd](mailto:aynal@eee.buet.ac.bd)

**Cell:** 018 1763 8600

## Topic 6

# Color Image Processing



# Introduction

Color image processing involves manipulating images with color information, using techniques like full-color processing (working directly with RGB/CMY) and pseudo-color processing (assigning colors to monochrome intensities) to enhance object recognition, improve quality, and extract features, often by converting between color models like RGB, CMYK, and HSI for analysis and transformation. It involves techniques for analyzing and enhancing digital images that contain color information, which provides richer data for object identification compared to grayscale.

# Importance

- **Human Perception:** Color aids object recognition and provides richer visual information than grayscale.
- **Enhanced Analysis:** Color adds vital information (chrominance) for detailed analysis in fields like medicine, remote sensing, and computer vision.
- **Automation:** Enables automated tasks like quality control, medical diagnostics, and content filtering.

# Processing Categories

Digital color image processing is primarily divided into two main areas:

- **Full-Color Processing:** Working with images captured by full-color sensors, such as digital cameras or scanners. These typically use 24-bit representation (8 bits per channel).
- **Pseudo-Color Processing:** Also known as **false-color processing**, this involves assigning colors to grayscale or monochrome intensities based on specific criteria to enhance visual interpretation.

# Color Models

Color models provide standard coordinate systems to specify and represent colors numerically.

- **RGB** (Red, Green, Blue): An additive model used in digital displays and cameras. Colors are formed by adding varying intensities of light.
- **CMY / CMYK** (Cyan, Magenta, Yellow, Black): A subtractive model used in printing. CMYK adds a black channel to improve quality and save ink.
- **HIS / HSV** (Hue, Saturation, Intensity/Value): A model that decouples color (Hue and Saturation) from brightness (Intensity/Value), making it highly effective for algorithms aligned with human perception.
- **YUV / YCbCr / YIQ**: Used in video and television broadcasting to separate luminance (Y) from chrominance (color), which allows for efficient data compression.
- **CIELAB** (Lab): A device-independent model designed to be perceptually uniform, meaning equal changes in values correspond to equal perceived changes in color.

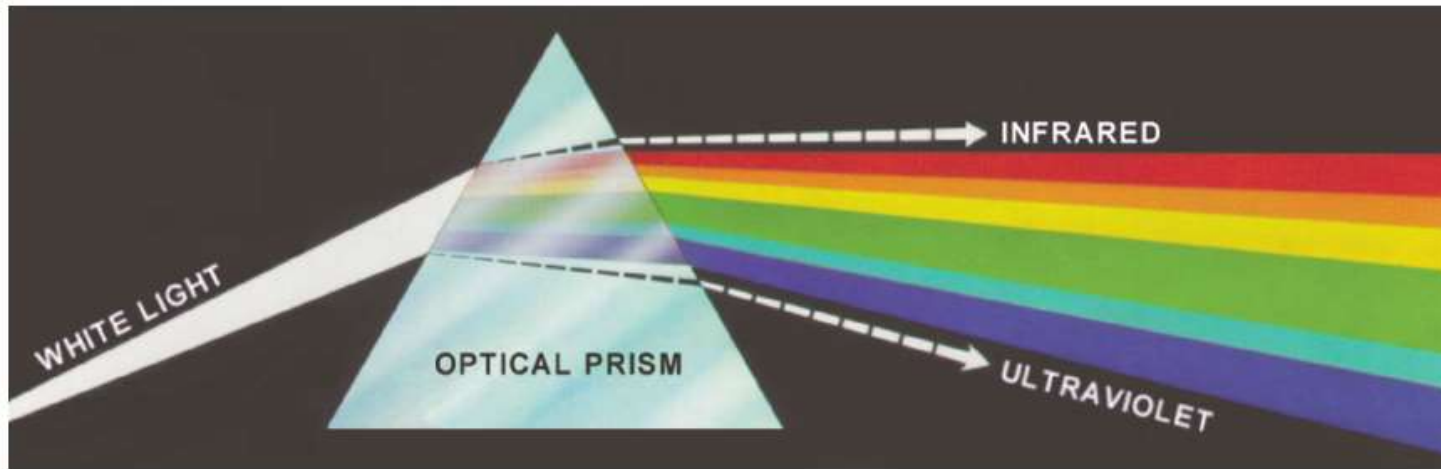
# Key Techniques

- **Color Transformations:** Converting images between different color spaces (e.g., RGB to HSI) to facilitate specific processing tasks.
- **Image Enhancement:** Includes contrast adjustment, **color balancing** (white balance), and histogram equalization.
- **Smoothing and Sharpening:** Smoothing reduces noise by averaging pixel values, while sharpening emphasizes edges and fine details.
- **Color Segmentation:** Partitioning an image into meaningful regions based on color similarities, often used for **object recognition**.
- **Noise Removal:** Techniques to clean random variations in color and intensity, such as Gaussian or salt-and-pepper noise, using specialized filters.

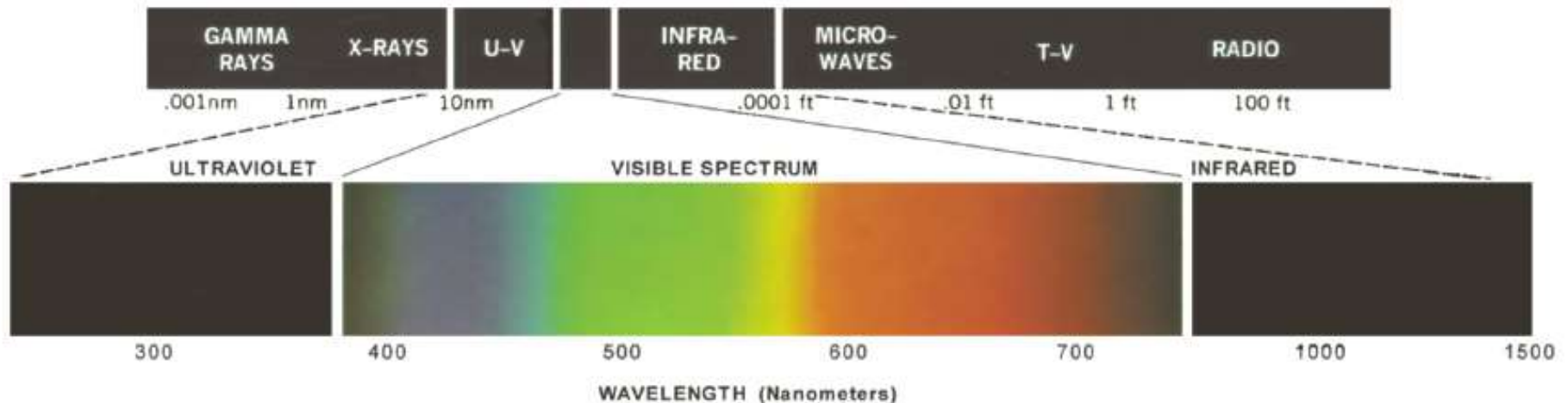
# Application Areas

- **Medical Imaging:** Highlighting specific tissues or abnormalities in MRI and CT scans.
- **Computer Vision:** Detecting objects, such as traffic signs or faces, by isolating specific color ranges.
- **Satellite Imagery:** Enhancing features in multispectral data for environmental monitoring.
- **Digital Restoration:** Restoring colors in old films or colorizing grayscale photography

# Color Fundamentals



**FIGURE 6.1** Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lamp Business Division.)



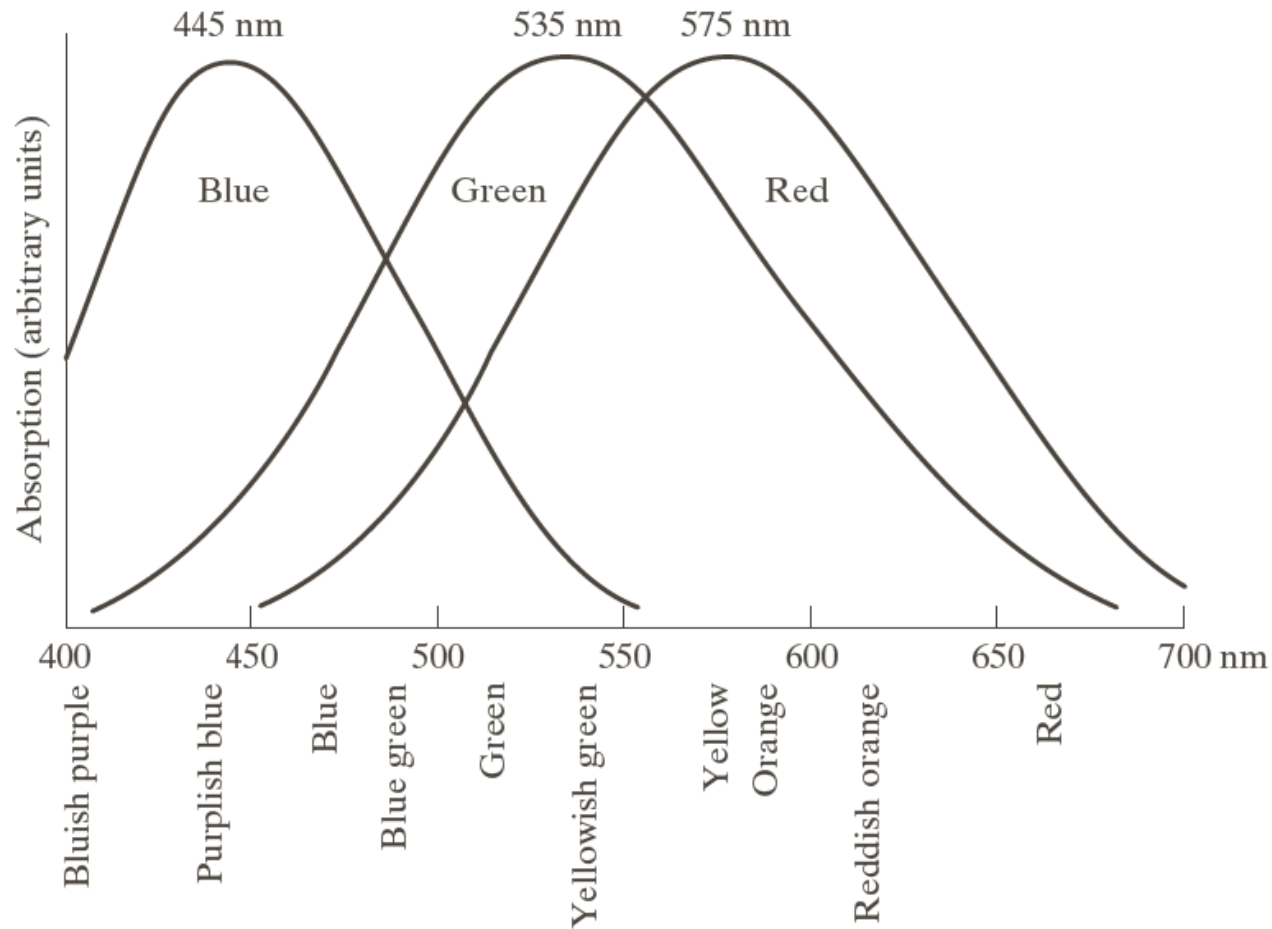
**FIGURE 6.2** Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

# Color Fundamentals

- ▶ 6 to 7 million cones in the human eye can be divided into three principal sensing categories, corresponding roughly to red, green, and blue.

65%: red    33%: green    2%: blue (blue cones are the most sensitive)

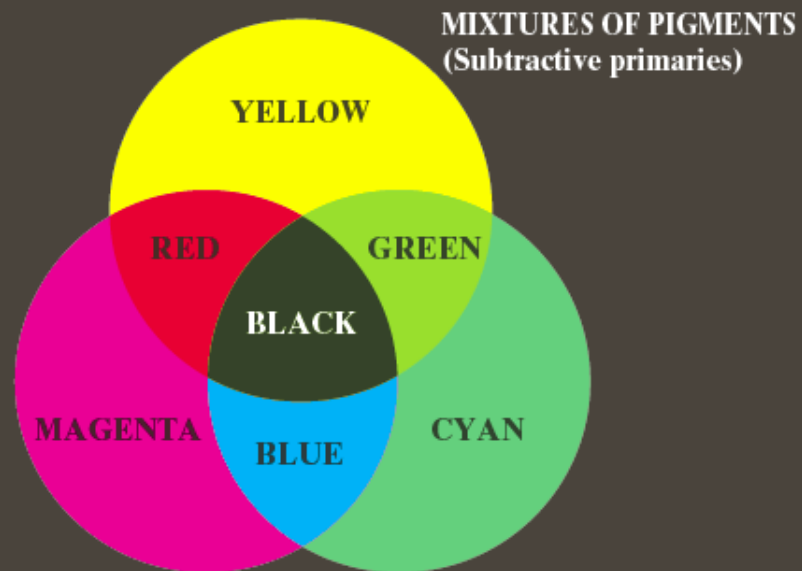
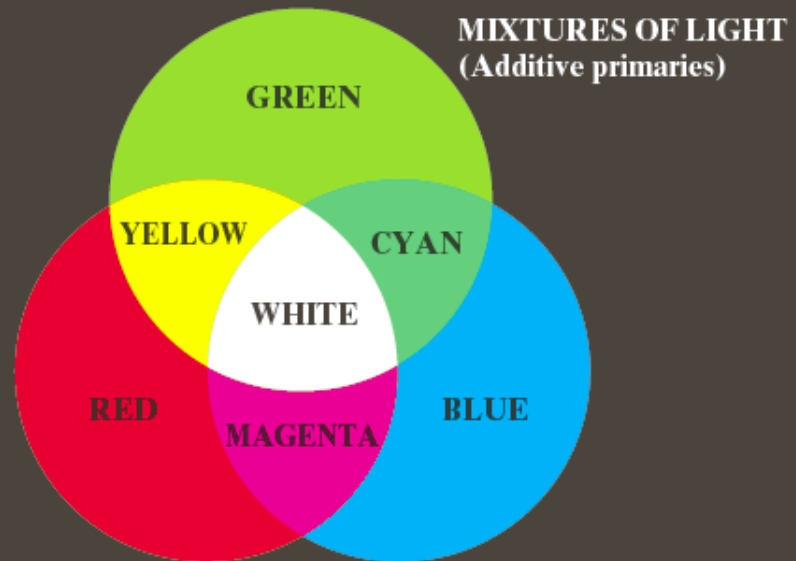
# Color Fundamentals



**FIGURE 6.3**  
Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

a  
b

**FIGURE 6.4**  
Primary and  
secondary colors  
of light and  
pigments.  
(Courtesy of the  
General Electric  
Co., Lamp  
Business  
Division.)



**PRIMARY AND SECONDARY COLORS  
OF LIGHT AND PIGMENT**

# Color Fundamentals

- ▶ The characteristics generally used to distinguish one color from another are brightness, hue, and saturation

**Brightness:** the achromatic notion of intensity.

**Hue:** dominant wavelength in a mixture of light waves, represents dominant color as perceived by an observer.

**Saturation:** relative purity or the amount of white light mixed with its hue.

# Color Fundamentals

## ► Tristimulus

Red, green, and blue are denoted X, Y, and Z, respectively. A color is defined by its trichromatic coefficients, defined as

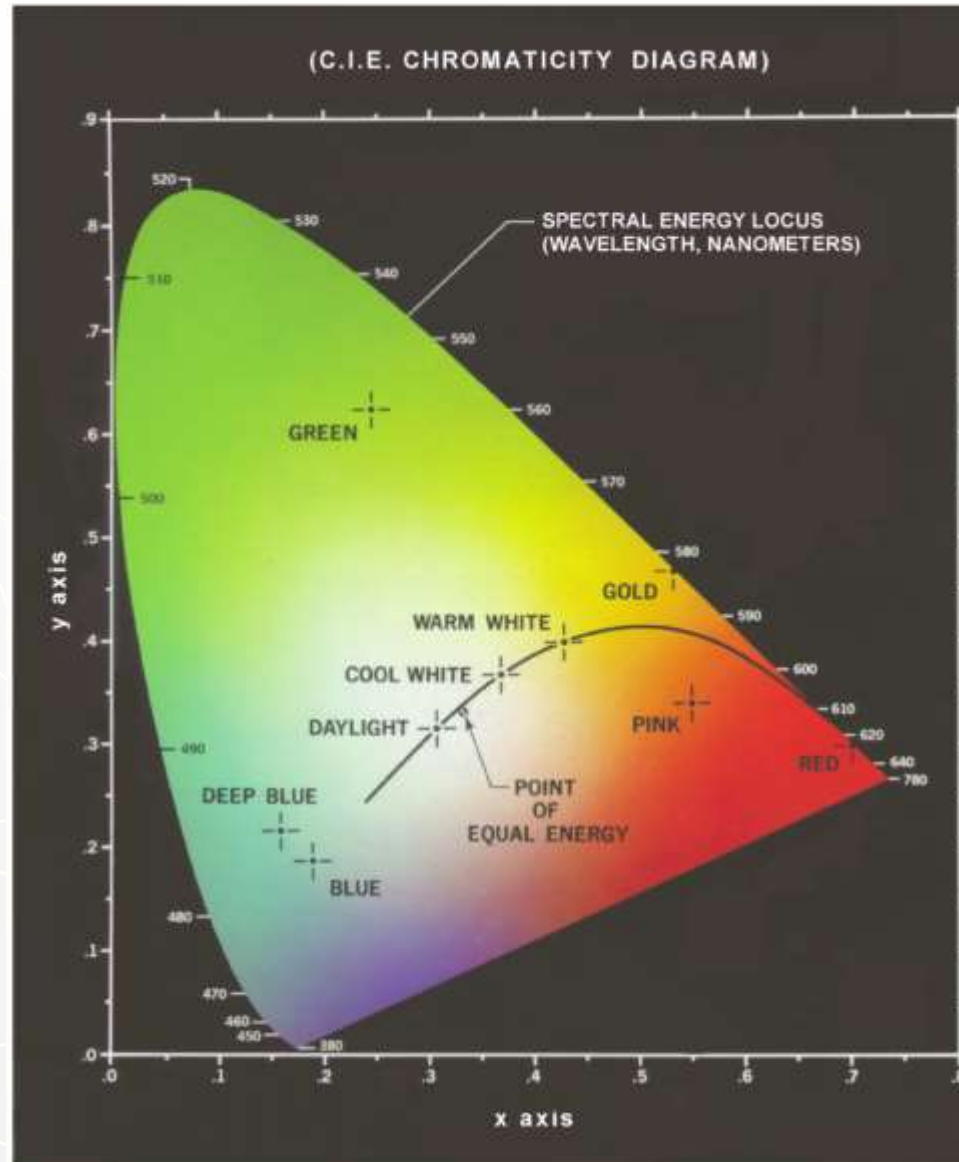
$$x = \frac{X}{X + Y + Z}$$

$$y = \frac{Y}{X + Y + Z}$$

$$z = \frac{Z}{X + Y + Z}$$

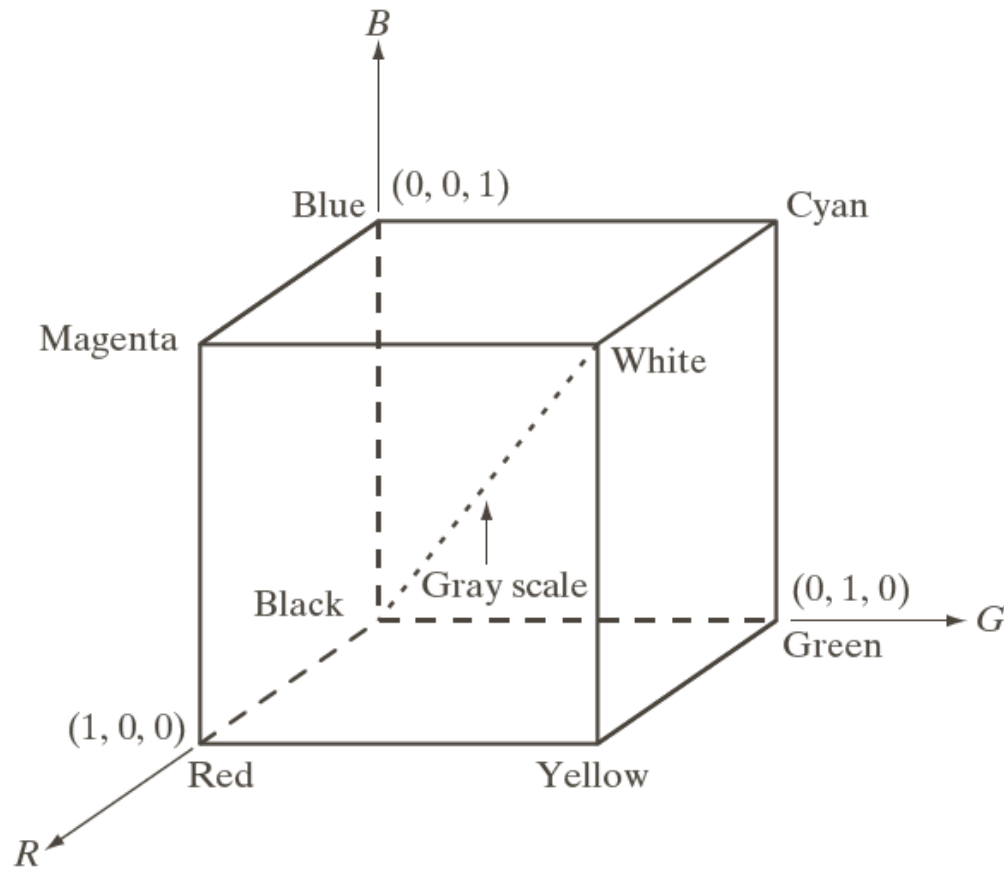
# CIE Chromaticity Diagram

It shows color composition as a function of x (red) and y (green)



**FIGURE 6.5**  
Chromaticity diagram.  
(Courtesy of the General Electric Co., Lamp Business Division.)

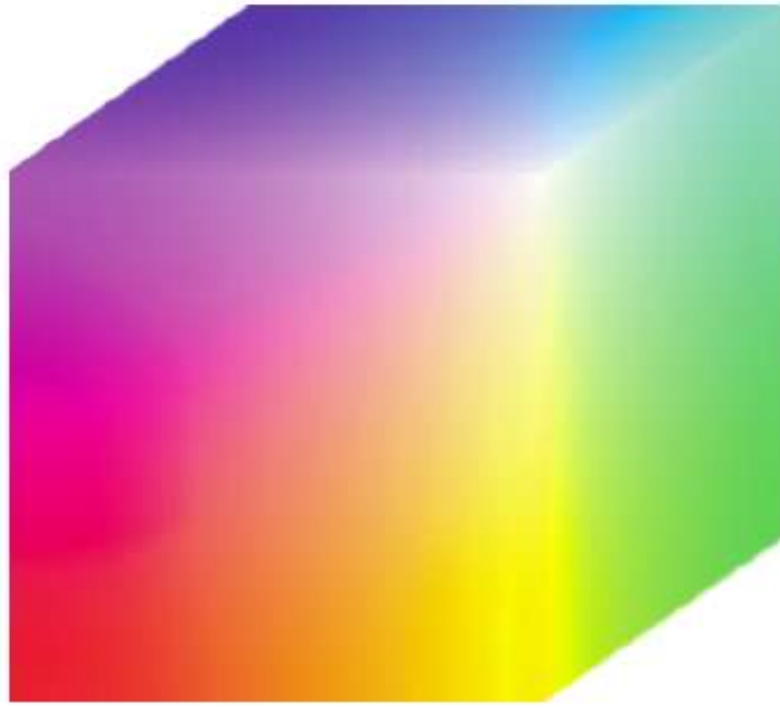
# RGB Color Model



**FIGURE 6.7**

Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point  $(1, 1, 1)$ .

# RGB Color Model



**FIGURE 6.8** RGB  
24-bit color cube.

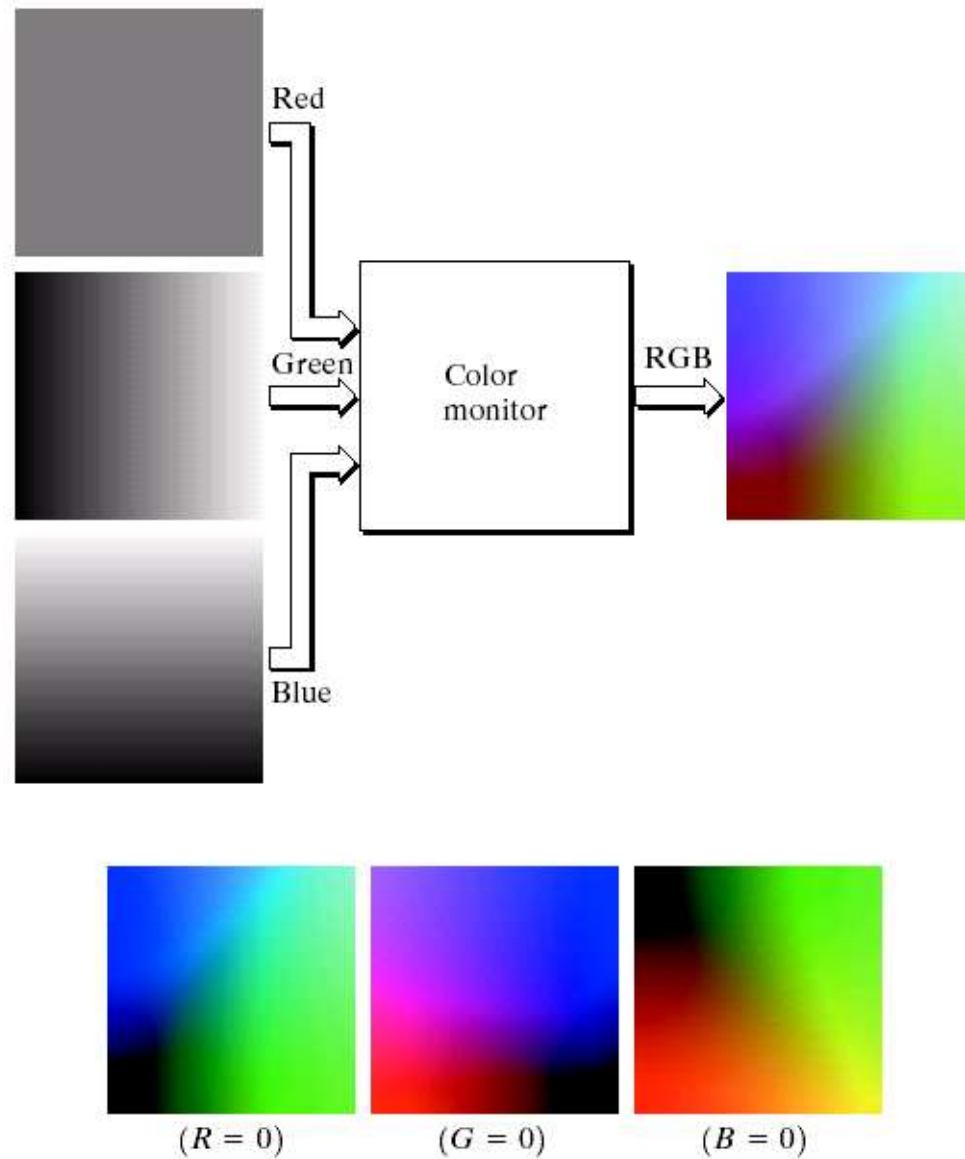
Pixel depth

The total number of colors  
in a 24-bit RGB image is  
 $(2^8)^3 = 16,777,216$

a  
b

**FIGURE 6.9**

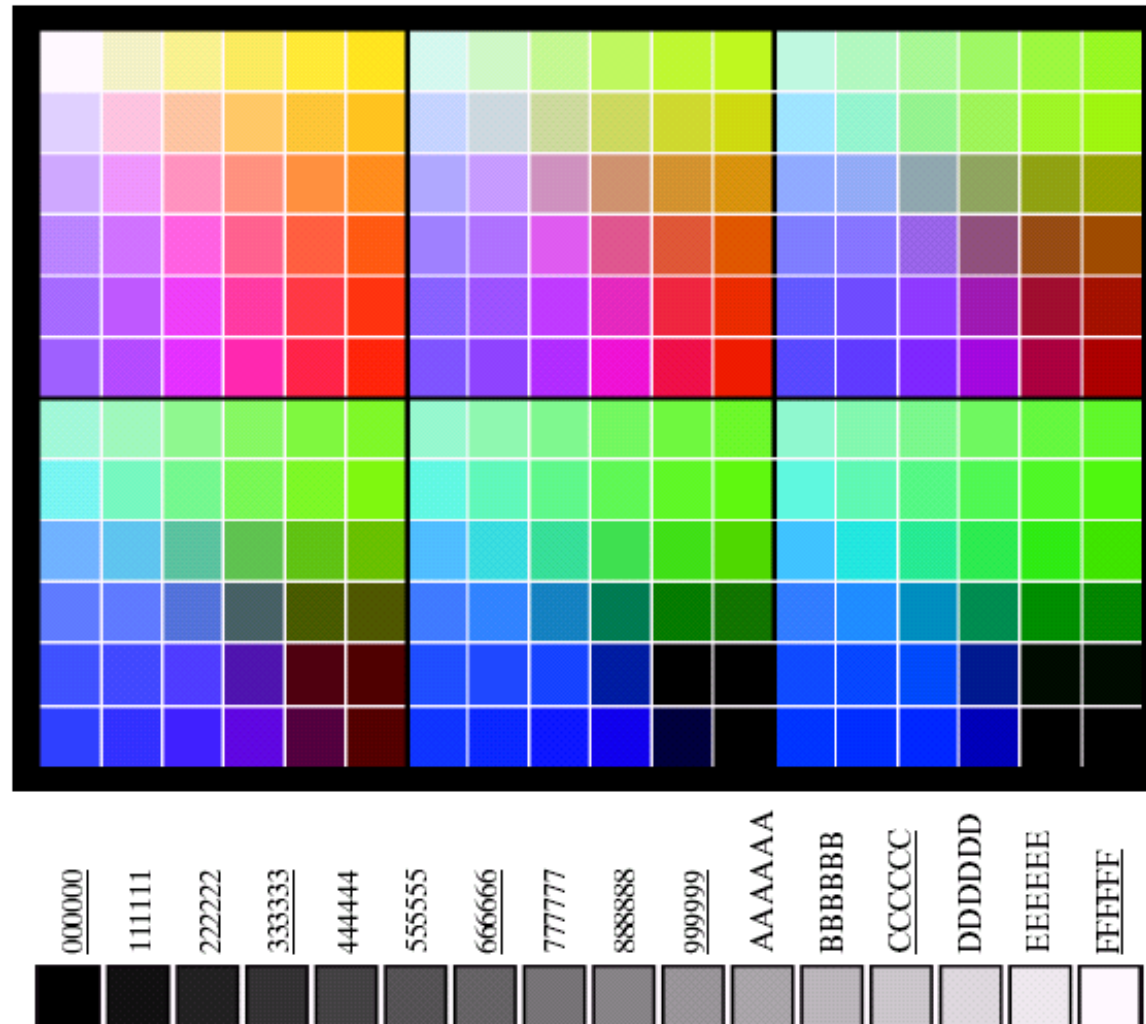
(a) Generating the RGB image of the cross-sectional color plane (127,  $G$ ,  $B$ ).  
(b) The three hidden surface planes in the color cube of Fig. 6.8.



| Number System |    | Color Equivalents |     |     |     |     |  |
|---------------|----|-------------------|-----|-----|-----|-----|--|
| Hex           | 00 | 33                | 66  | 99  | CC  | FF  |  |
| Decimal       | 0  | 51                | 102 | 153 | 204 | 255 |  |

**TABLE 6.1**

Valid values of each RGB component in a safe color.

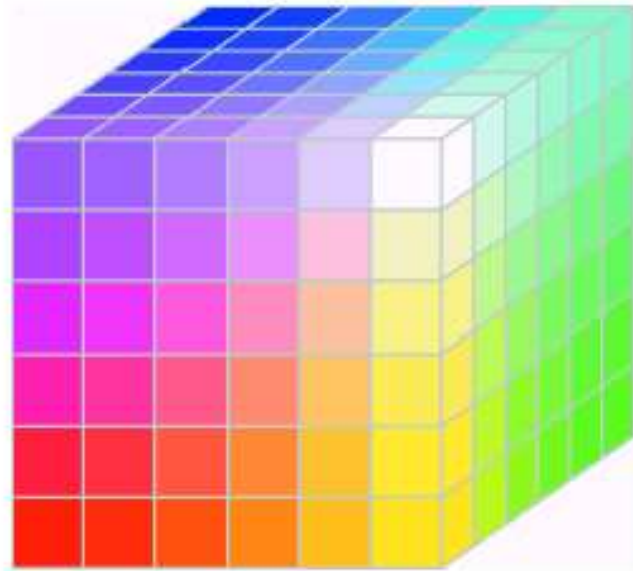


a  
b

**FIGURE 6.10**

(a) The 216 safe RGB colors.  
(b) All the grays in the 256-color RGB system (grays that are part of the safe color group are shown underlined).

**Safe RGB colors** (or **safe Web colors**) are reproduced faithfully, reasonably independently of viewer hardware capabilities



**FIGURE 6.11** The RGB safe-color cube.

# The CMY and CMYK Color Models

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Equal amounts of the pigment primaries, cyan, magenta, and yellow should produce black. In practice, combining these colors for printing produces a muddy-looking black.

To produce true black, the predominant color in printing, the fourth color, black, is added, giving rise to the CMYK color model.



CMY vs. CMYK



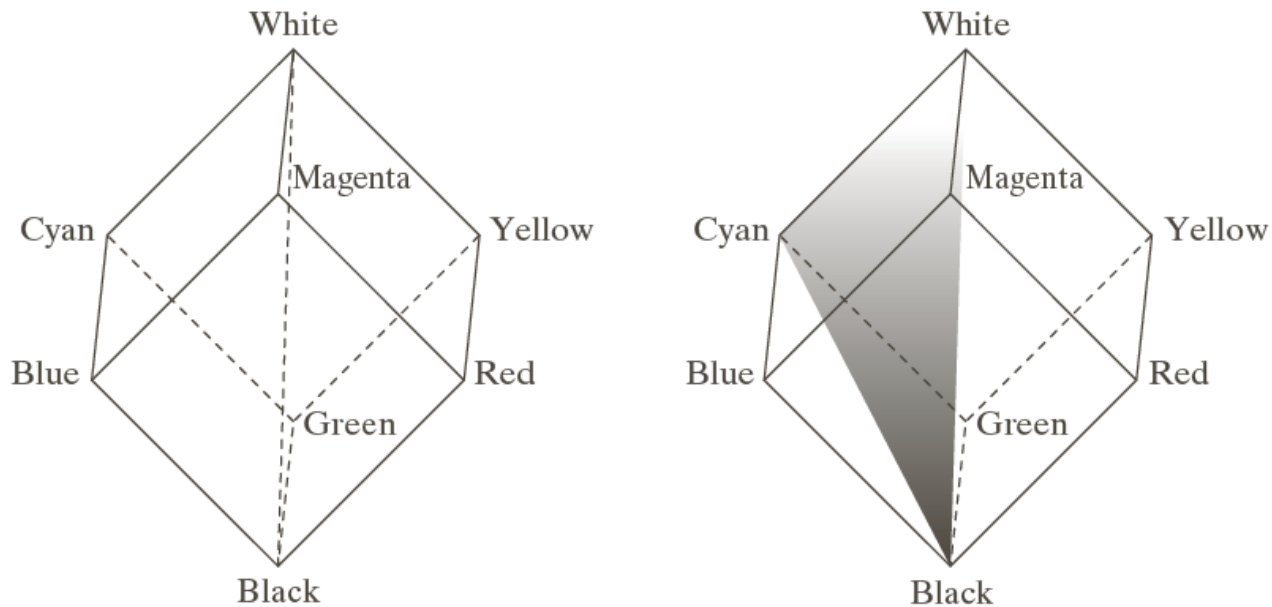
# HSI Color Model

**brightness:** the achromatic notion of **intensity**.

**hue:** dominant wavelength in a mixture of light waves, represents dominant color as perceived by an observer.

**saturation:** relative purity or the amount of white light mixed with its hue.

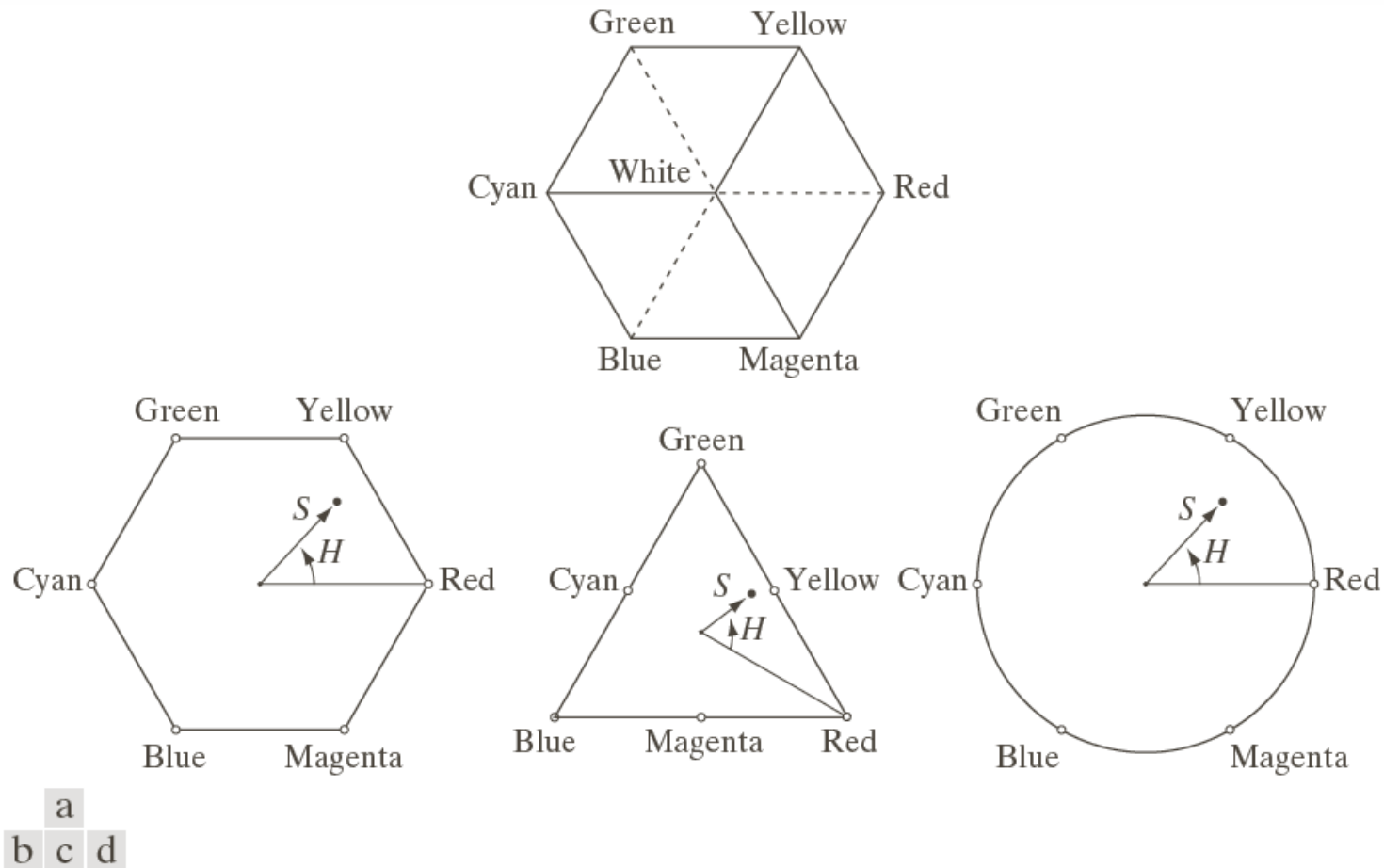
# HSI Color Model



a b

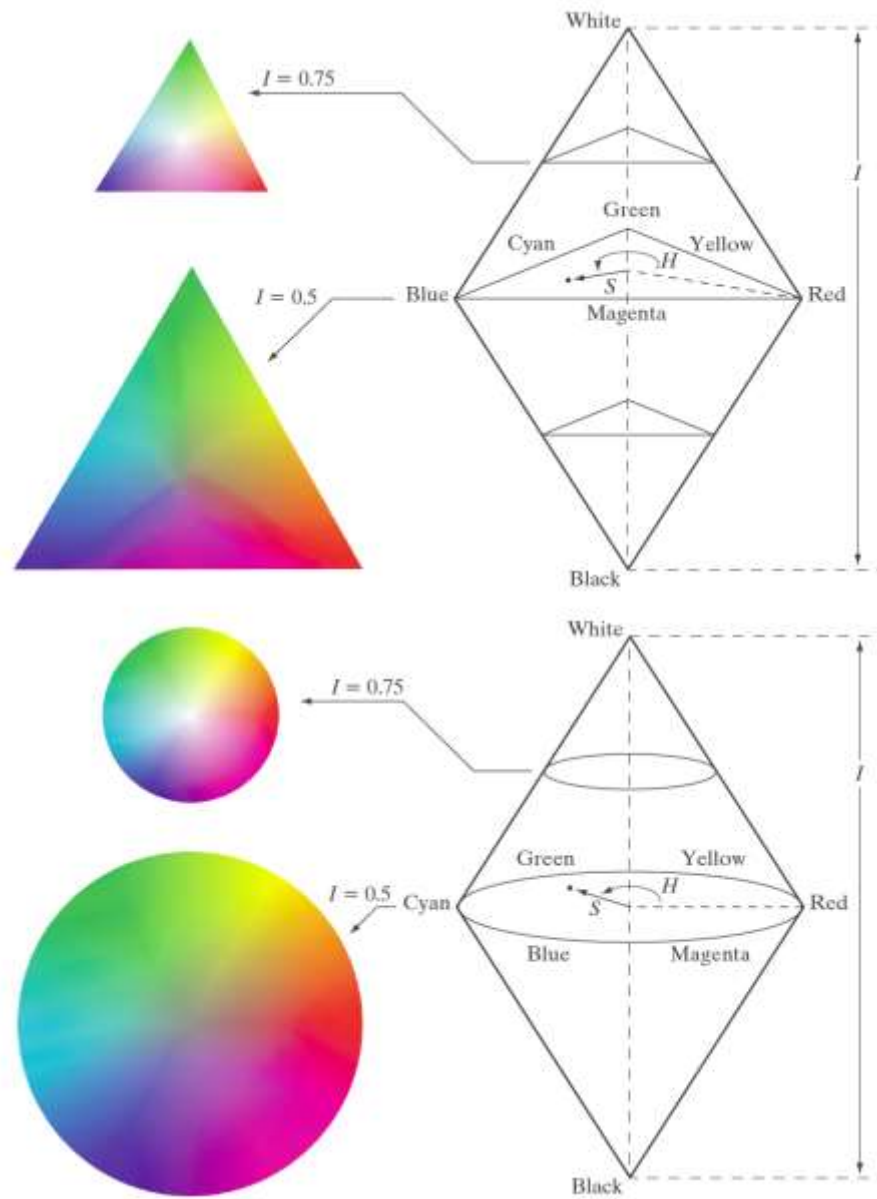
**FIGURE 6.12**  
Conceptual  
relationships  
between the RGB  
and HSI color  
models.

# HSI Color Model



**FIGURE 6.13** Hue and saturation in the HSI color model. The dot is an arbitrary color point. The angle from the red axis gives the hue, and the length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

# HSI Color Model



a  
b

**FIGURE 6.14** The HSI color model based on (a) triangular and (b) circular color planes. The triangles and circles are perpendicular to the vertical intensity axis.

# Converting Colors from RGB to HSI

- ▶ Given an image in RGB color format, the H component of each RGB pixel is obtained using the equation

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R-G) + (R-B)]}{\left[ (R-G)^2 + (R-B)(G-B) \right]^{1/2}} \right\}$$

# Converting Colors from RGB to HSI

- ▶ Given an image in RGB color format, the saturation component is given by

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

The intensity component is given by

$$I = \frac{1}{3}(R + G + B)$$

# Converting Colors from HSI to RGB

► RG sector  $(0^\circ \leq H < 120^\circ)$

$$B = I(1 - S)$$

$$R = I \left[ 1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

and

$$G = 3I - (R + B)$$

# Converting Colors from HSI to RGB

► RG sector ( $120^\circ \leq H < 240^\circ$ )

$$H = H - 120^\circ$$

$$R = I(1 - S)$$

$$G = I \left[ 1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

and

$$B = 3I - (R + G)$$

# Converting Colors from HSI to RGB

► RG sector ( $240^\circ \leq H \leq 360^\circ$ )

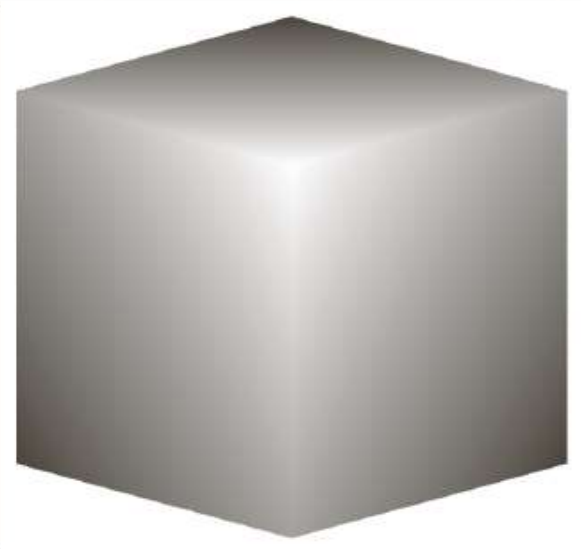
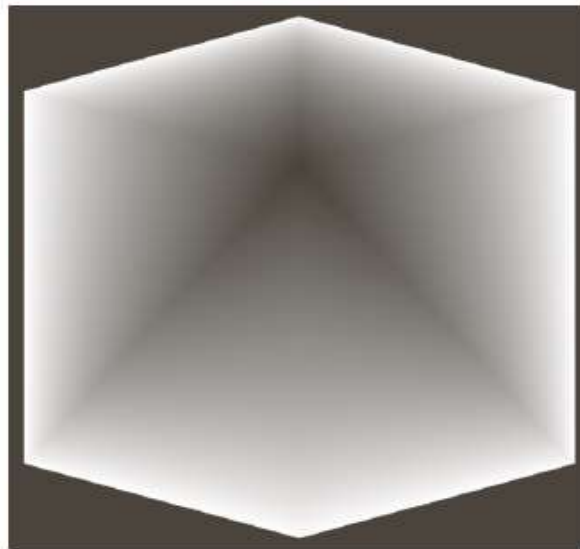
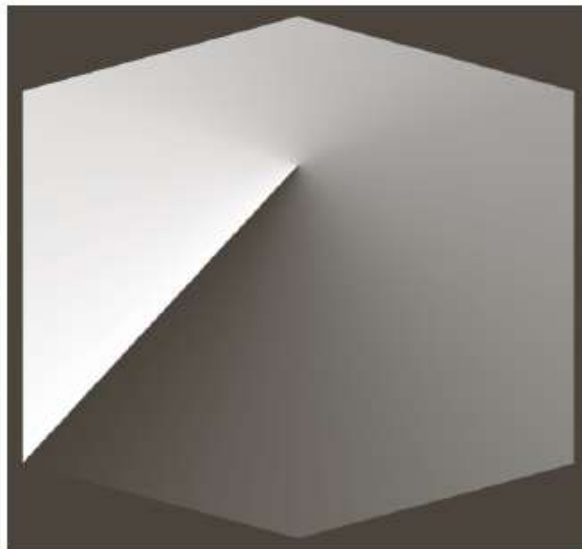
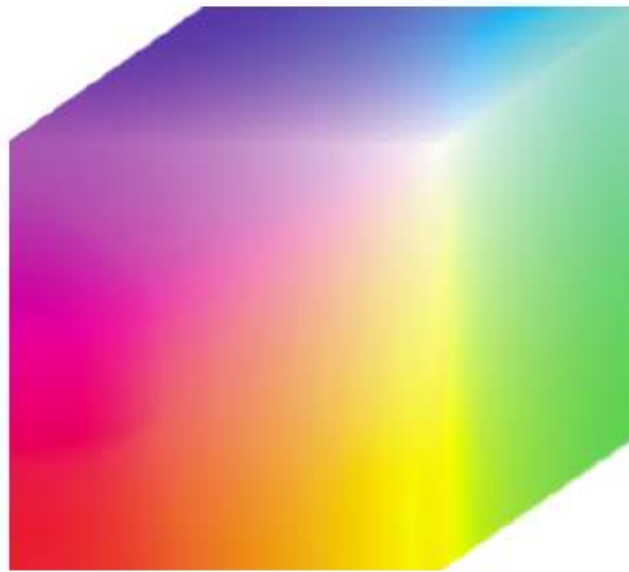
$$H = H - 240^\circ$$

$$G = I(1 - S)$$

$$B = I \left[ 1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

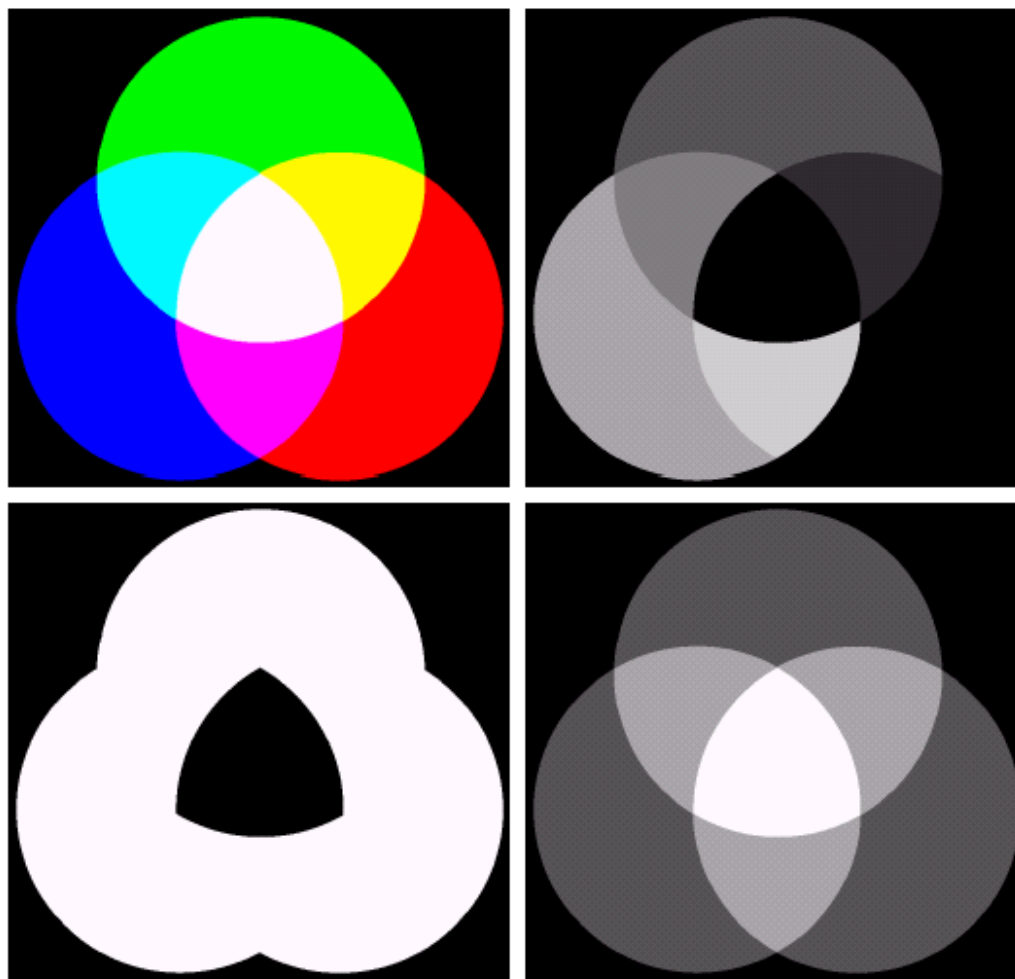
and

$$R = 3I - (G + B)$$



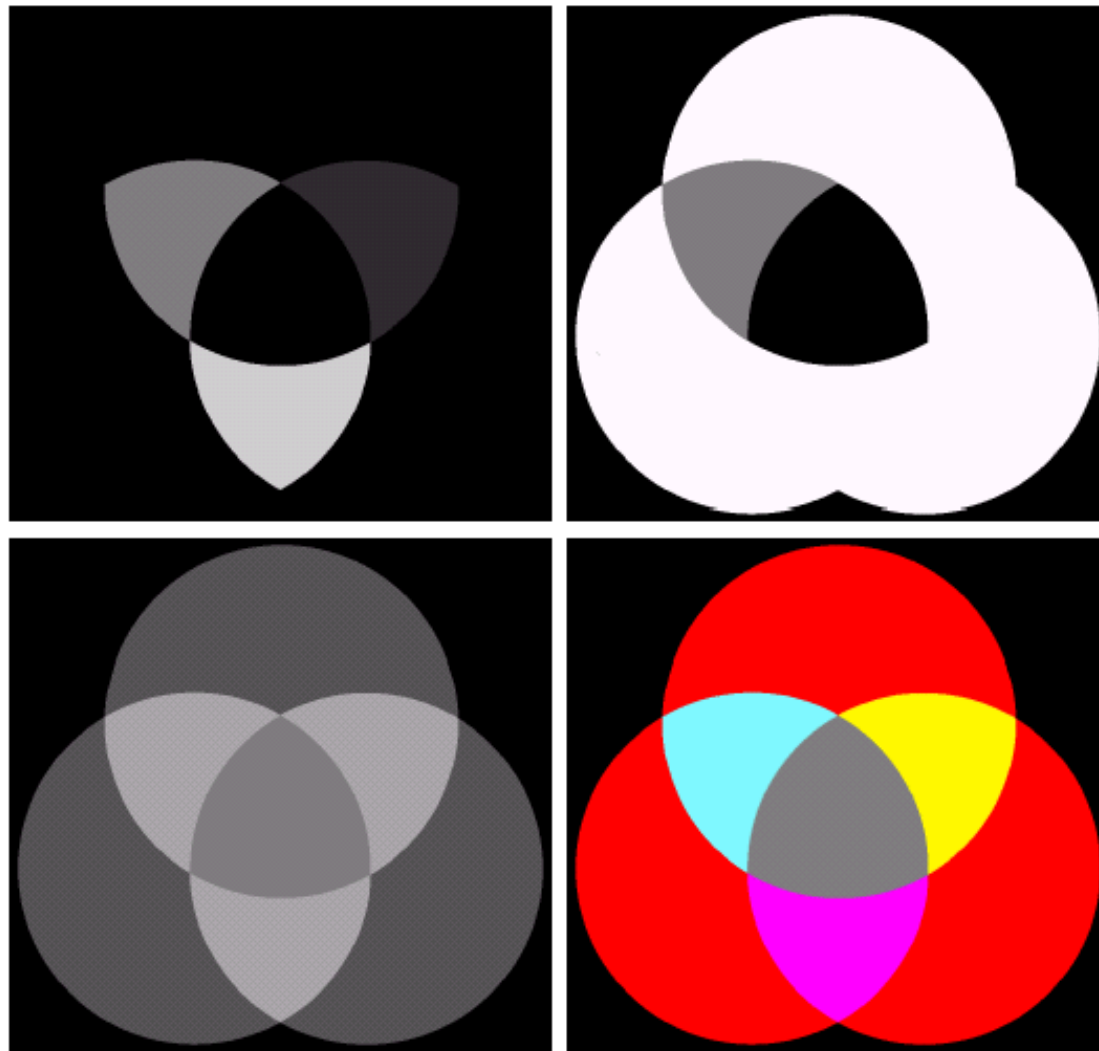
a b c

**FIGURE 6.15** HSI components of the image in Fig. 6.8. (a) Hue, (b) saturation, and (c) intensity images.



a b  
c d

**FIGURE 6.16** (a) RGB image and the components of its corresponding HSI image: (b) hue, (c) saturation, and (d) intensity.



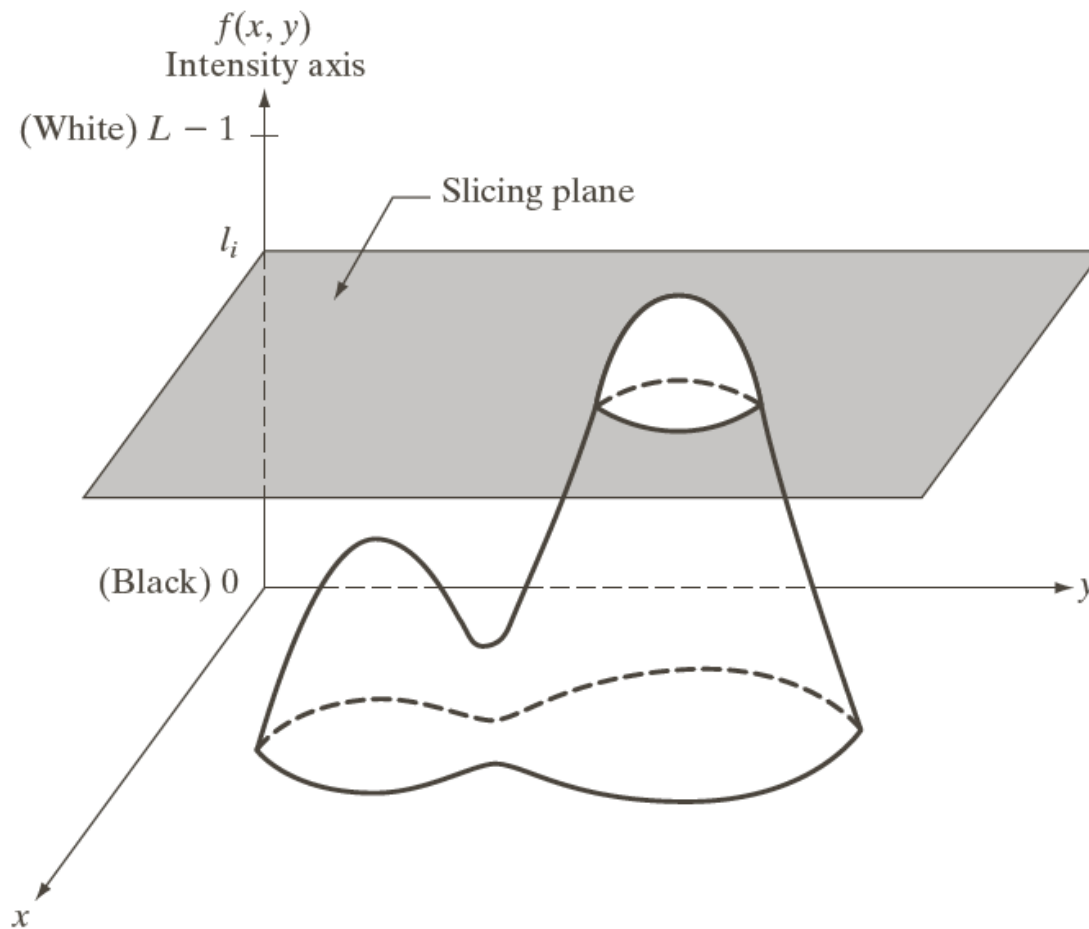
|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 6.17** (a)–(c) Modified HSI component images. (d) Resulting RGB image. (See Fig. 6.16 for the original HSI images.)

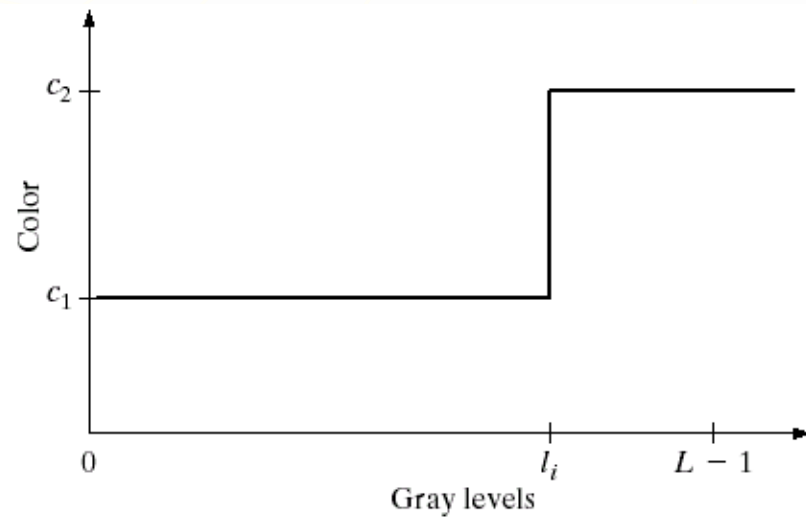
# Pseudocolor Image Processing

- ▶ The process of assigning colors to gray values based on a specified criterion.
- ▶ Intensity Slicing

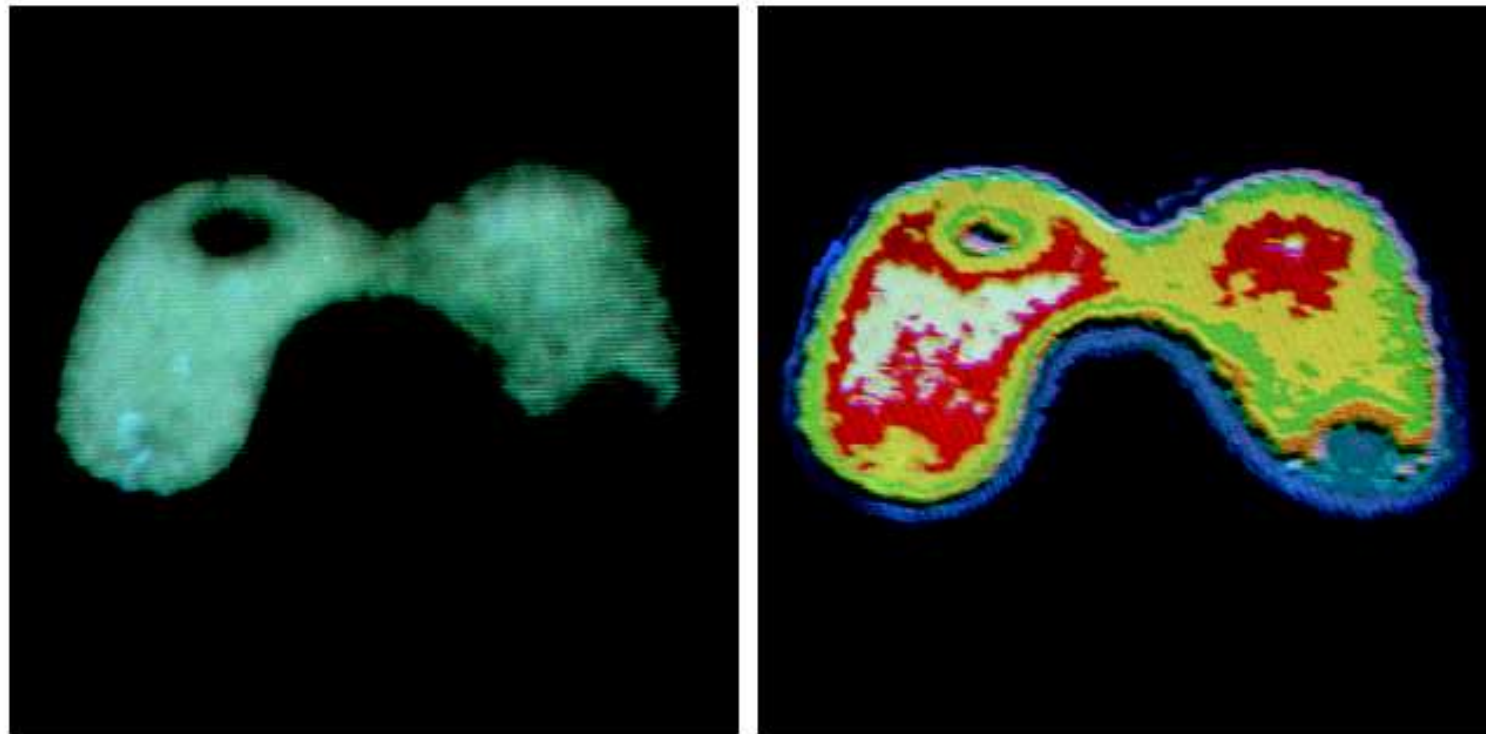
$$f(x, y) = c_k \quad \text{if } f(x, y) \in V_k$$



**FIGURE 6.18**  
 Geometric  
 interpretation of  
 the intensity-  
 slicing technique.



**FIGURE 6.19** An alternative representation of the intensity-slicing technique.



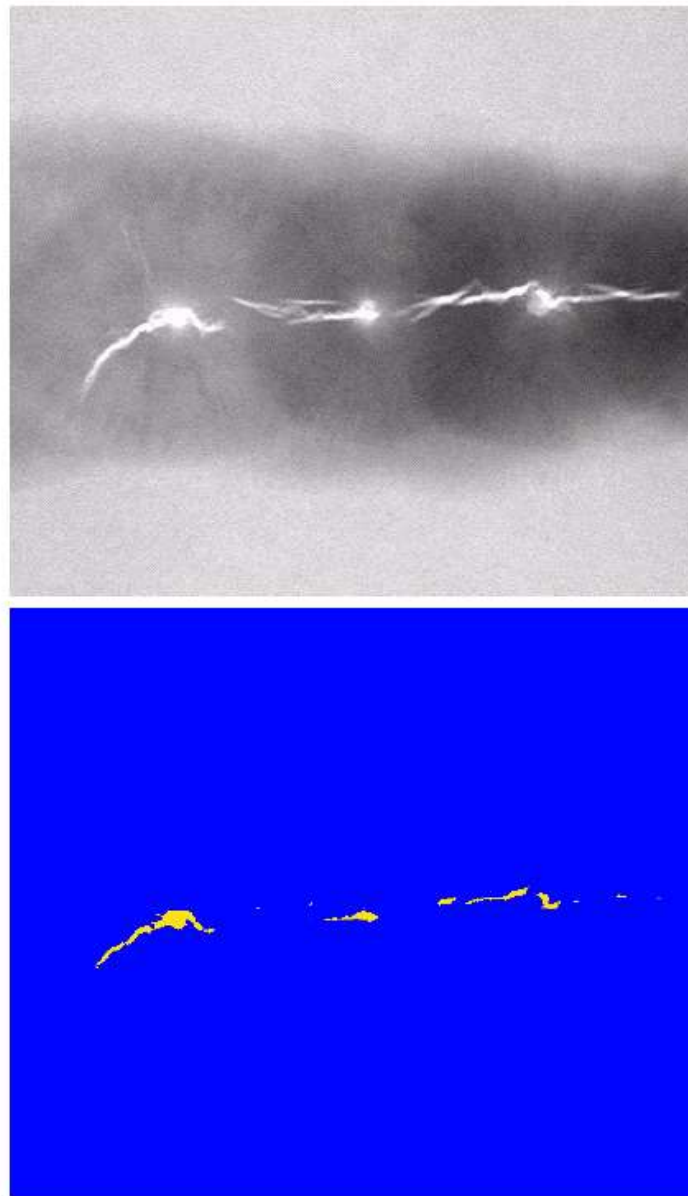
a b

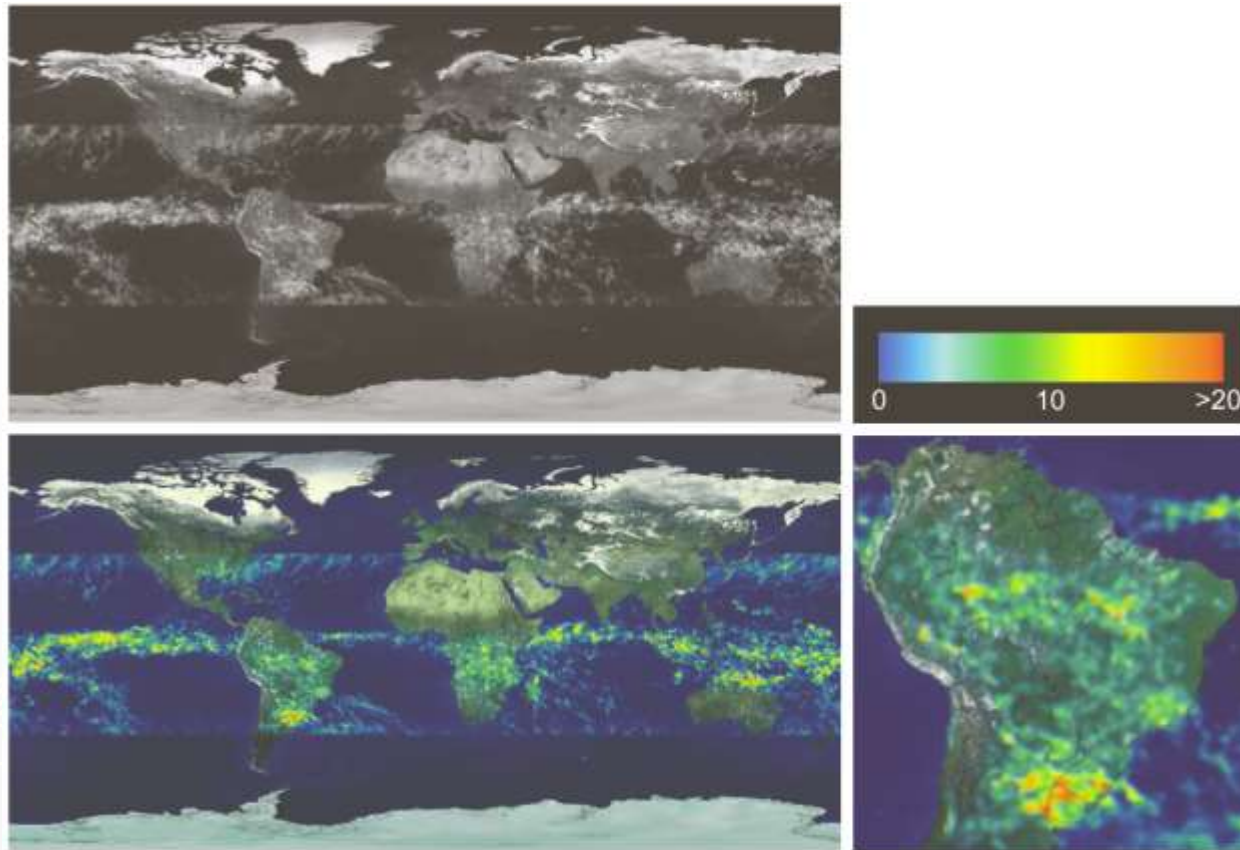
**FIGURE 6.20** (a) Monochrome image of the Picker Thyroid Phantom. (b) Result of density slicing into eight colors. (Courtesy of Dr. J. L. Blankenship, Instrumentation and Controls Division, Oak Ridge National Laboratory.)

a  
b

**FIGURE 6.21**

(a) Monochrome X-ray image of a weld. (b) Result of color coding. (Original image courtesy of X-TEK Systems, Ltd.)



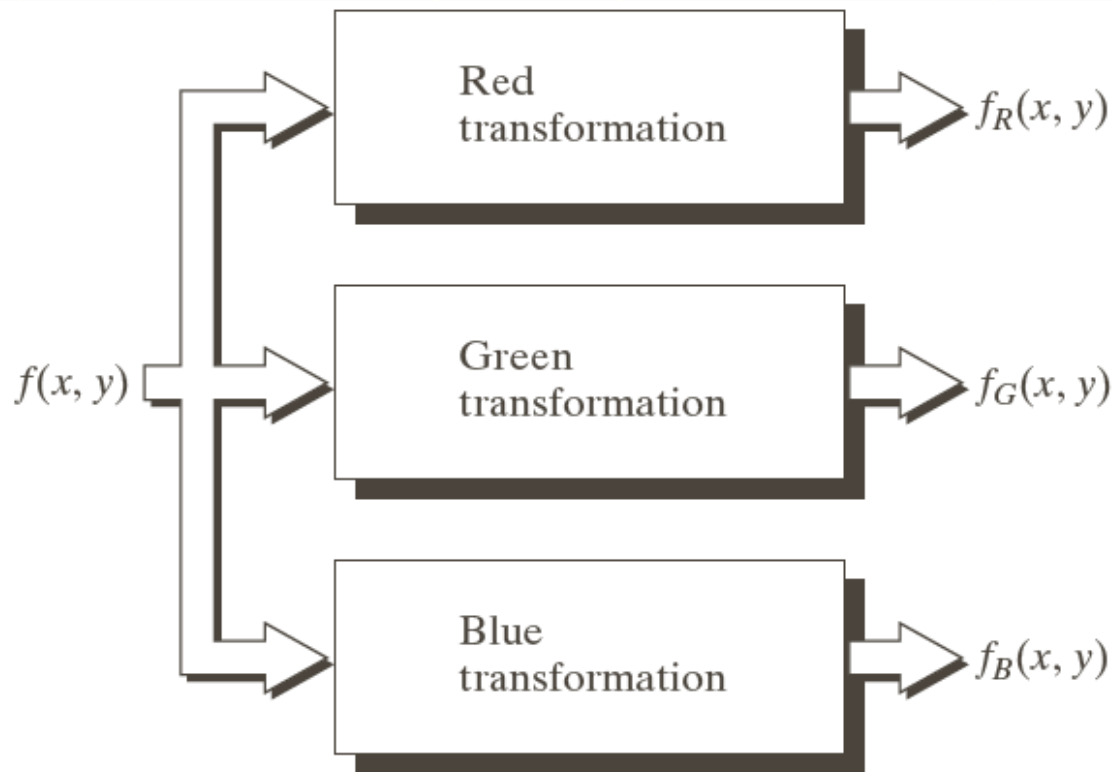


|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 6.22** (a) Gray-scale image in which intensity (in the lighter horizontal band shown) corresponds to average monthly rainfall. (b) Colors assigned to intensity values. (c) Color-coded image. (d) Zoom of the South American region. (Courtesy of NASA.)

# Pseudocolor Image Processing

## ► Intensity to Color Transformation

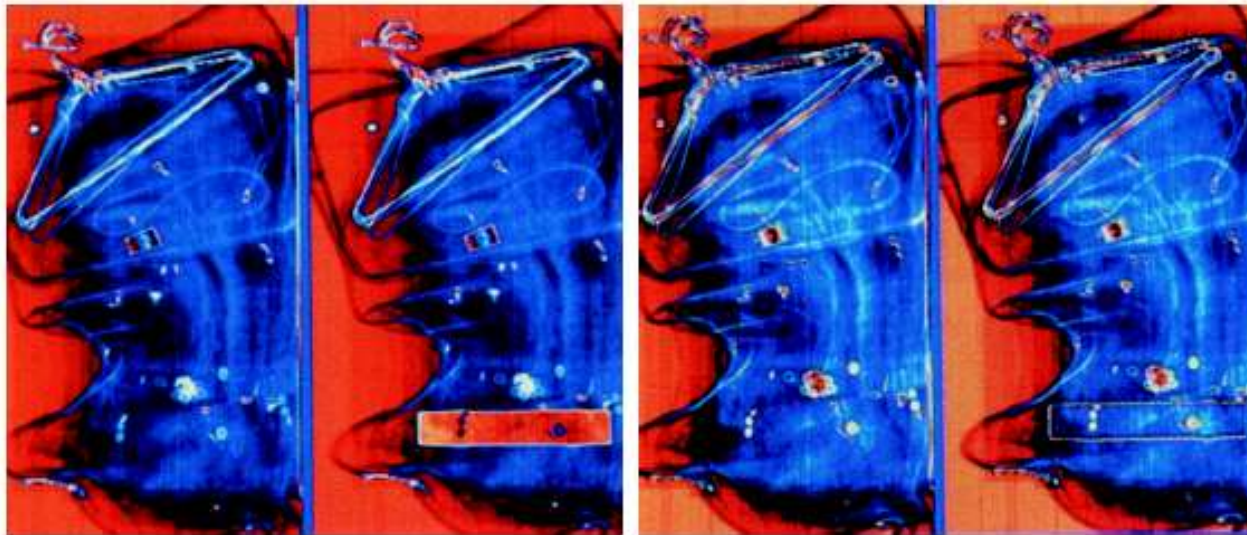
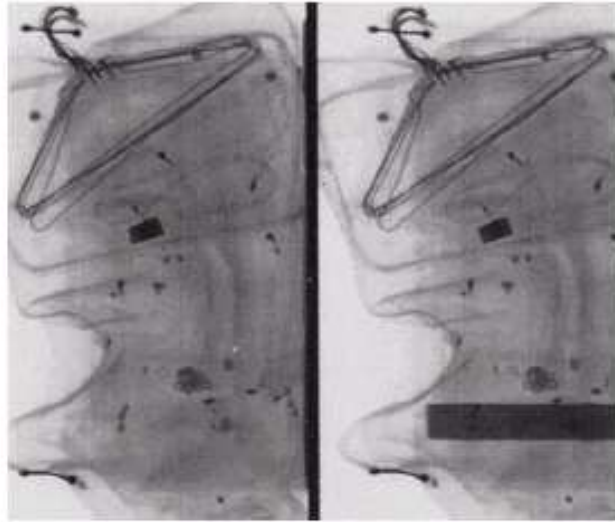


**FIGURE 6.23**

Functional block diagram for pseudocolor image processing.  $f_R$ ,  $f_G$ , and  $f_B$  are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

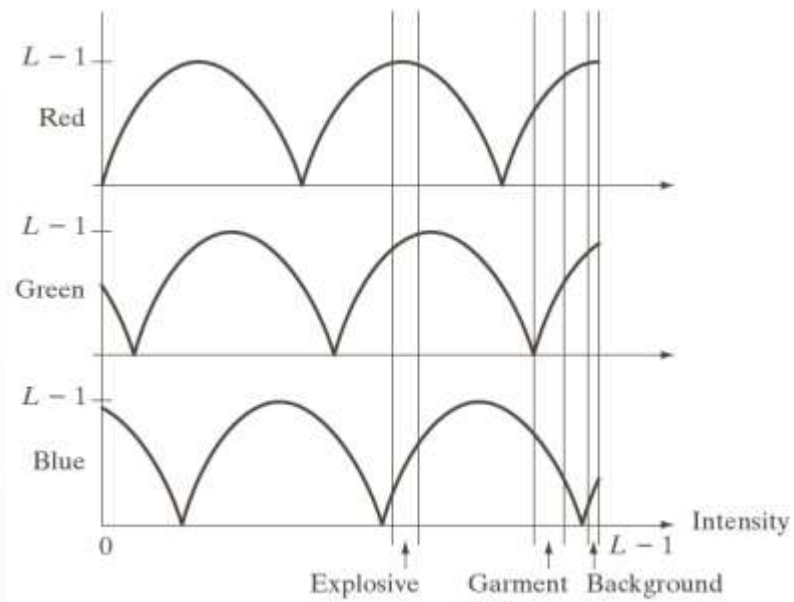
The images are obtained from an airport X-ray scanning system.

The left contains ordinary articles and the right contains the same articles as well as a block of simulated plastic explosives.



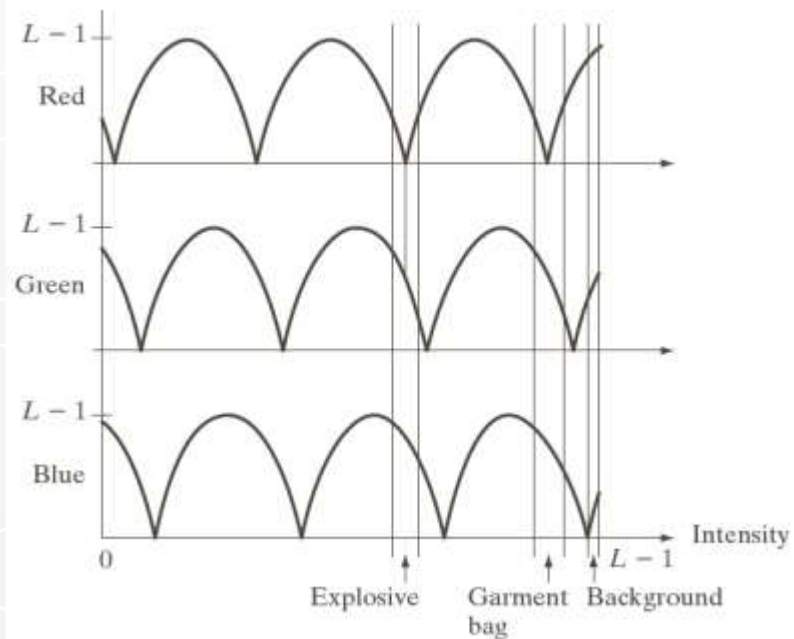
a  
b c

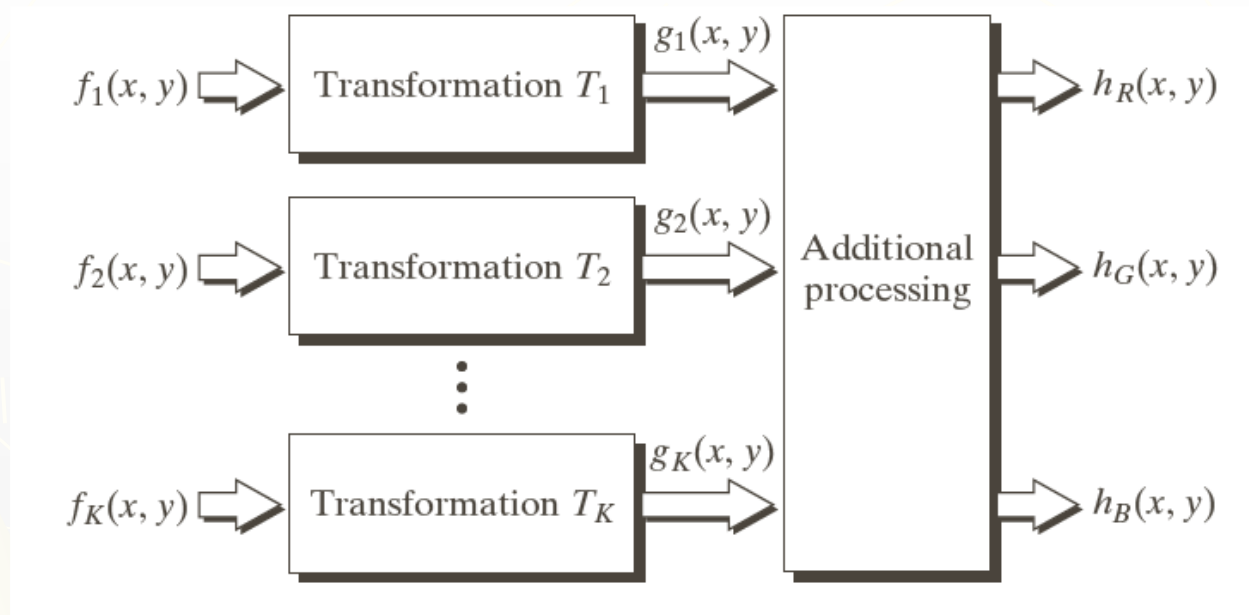
**FIGURE 6.24** Pseudocolor enhancement by using the gray-level to color transformations in Fig. 6.25. (Original image courtesy of Dr. Mike Hurwitz, Westinghouse.)



a  
b

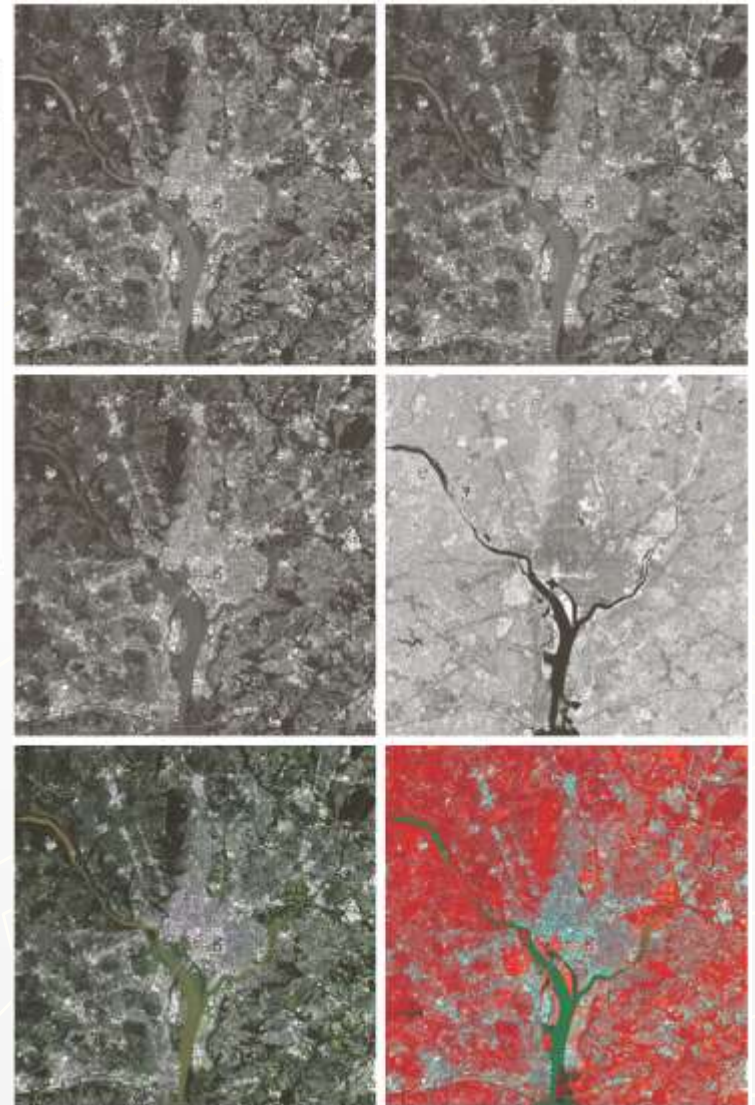
**FIGURE 6.25**  
Transformation  
functions used to  
obtain the images  
in Fig. 6.24.





**FIGURE 6.26** A pseudocolor coding approach used when several monochrome images are available.

| Band No. | Name             | Wavelength ( $\mu\text{m}$ ) | Characteristics and Uses                |
|----------|------------------|------------------------------|-----------------------------------------|
| 1        | Visible blue     | 0.45–0.52                    | Maximum water penetration               |
| 2        | Visible green    | 0.52–0.60                    | Good for measuring plant vigor          |
| 3        | Visible red      | 0.63–0.69                    | Vegetation discrimination               |
| 4        | Near infrared    | 0.76–0.90                    | Biomass and shoreline mapping           |
| 5        | Middle infrared  | 1.55–1.75                    | Moisture content of soil and vegetation |
| 6        | Thermal infrared | 10.4–12.5                    | Soil moisture; thermal mapping          |
| 7        | Middle infrared  | 2.08–2.35                    | Mineral mapping                         |



**FIGURE 6.27** (a)–(d) Images in bands 1–4 in Fig. 1.10 (see Table 1.1). (e) Color composite image obtained by treating (a), (b), and (c) as the red, green, blue components of an RGB image. (f) Image obtained in the same manner, but using in the red channel the near-infrared image in (d). (Original multispectral images courtesy of NASA.)

|   |   |
|---|---|
| a | b |
| c | d |
| e | f |

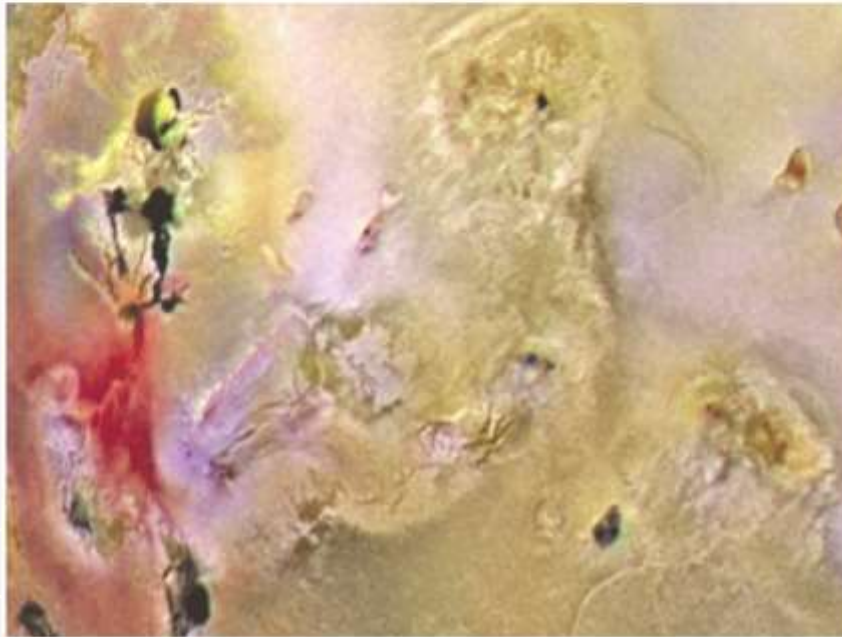
Pseudocolor by combining several of the sensor images from the Galileo spacecraft, some of which are in spectral regions not visible to the eye.

Bright red depicts materials newly ejected from an active volcano on Io, and the surrounding yellow materials are older sulfur deposits.



a  
b

**FIGURE 6.28**  
(a) Pseudocolor  
rendition of  
Jupiter Moon Io.  
(b) A close-up.  
(Courtesy of  
NASA.)



# Basics of Full-Color Image Processing

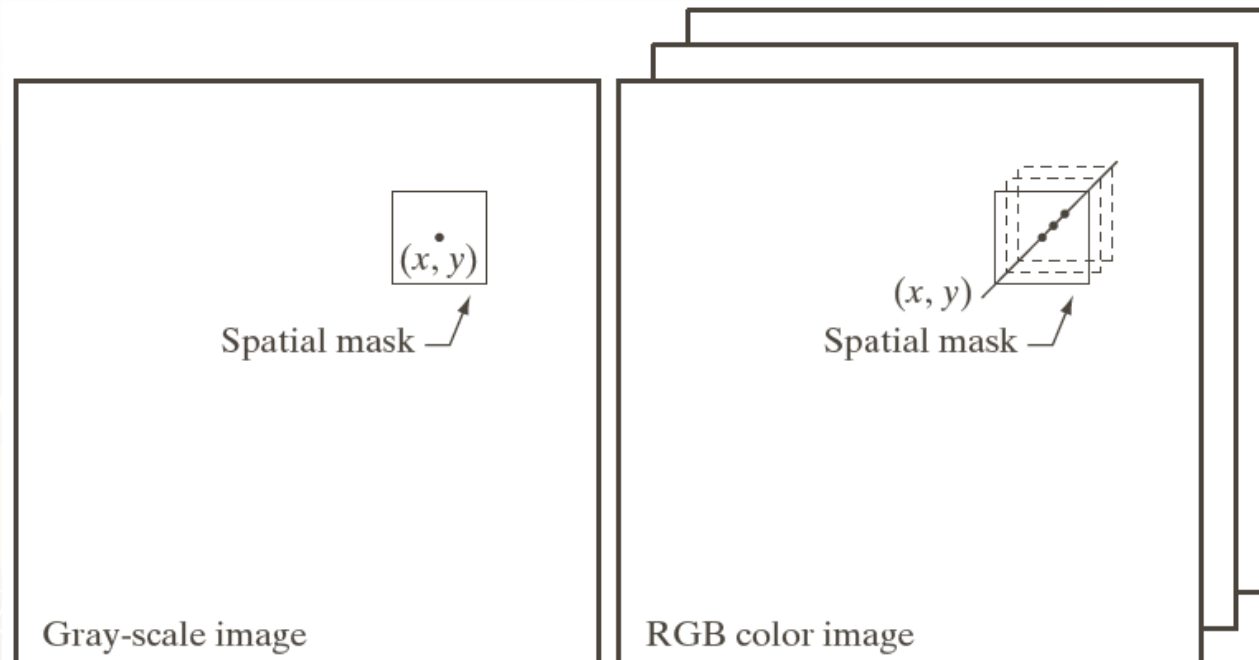
Let  $c$  represent an arbitrary vector in RGB color space:

$$c = \begin{bmatrix} c_R \\ c_G \\ c_B \end{bmatrix} = \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

At coordinates  $(x, y)$ ,

$$c(x, y) = \begin{bmatrix} c_R(x, y) \\ c_G(x, y) \\ c_B(x, y) \end{bmatrix} = \begin{bmatrix} R(x, y) \\ G(x, y) \\ B(x, y) \end{bmatrix}$$

# Basics of Full-Color Image Processing



a b

**FIGURE 6.29**  
Spatial masks for  
gray-scale and  
RGB color  
images.

# Color Transformations

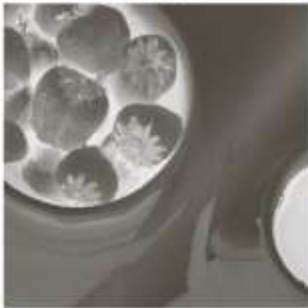
$$g(x, y) = T[f(x, y)]$$

$$s_i = T_i(r_1, r_2, \dots, r_n), \quad i = 1, 2, \dots, n.$$



Full color

**FIGURE 6.30** A full-color image and its various color-space components. (Interactive.)



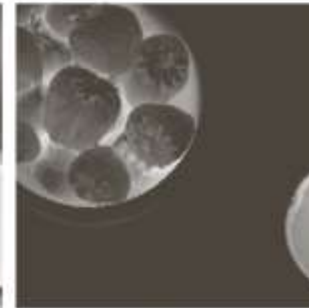
Cyan



Magenta



Yellow



Black



Red



Green



Blue



Hue



Saturation



Intensity

$$g(x, y) = kf(x, y)$$

|   |     |
|---|-----|
| a | b   |
| c | d e |

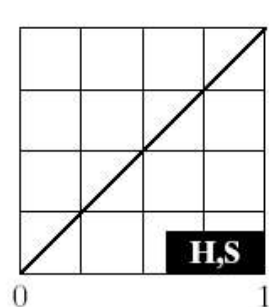
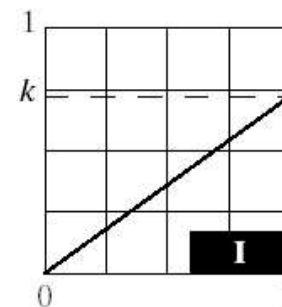
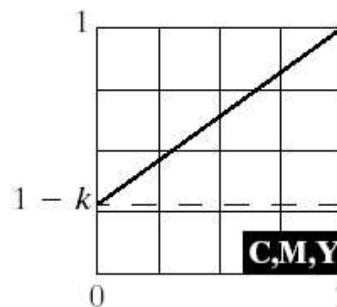
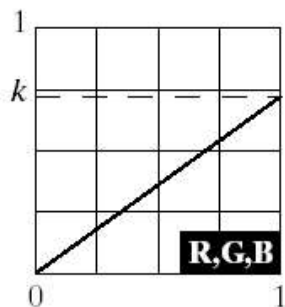
**FIGURE 6.31**

Adjusting the intensity of an image using color transformations.

(a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting  $k = 0.7$ ).

(c)–(e) The required RGB, CMY, and HSI transformation functions.

(Original image courtesy of MedData Interactive.)



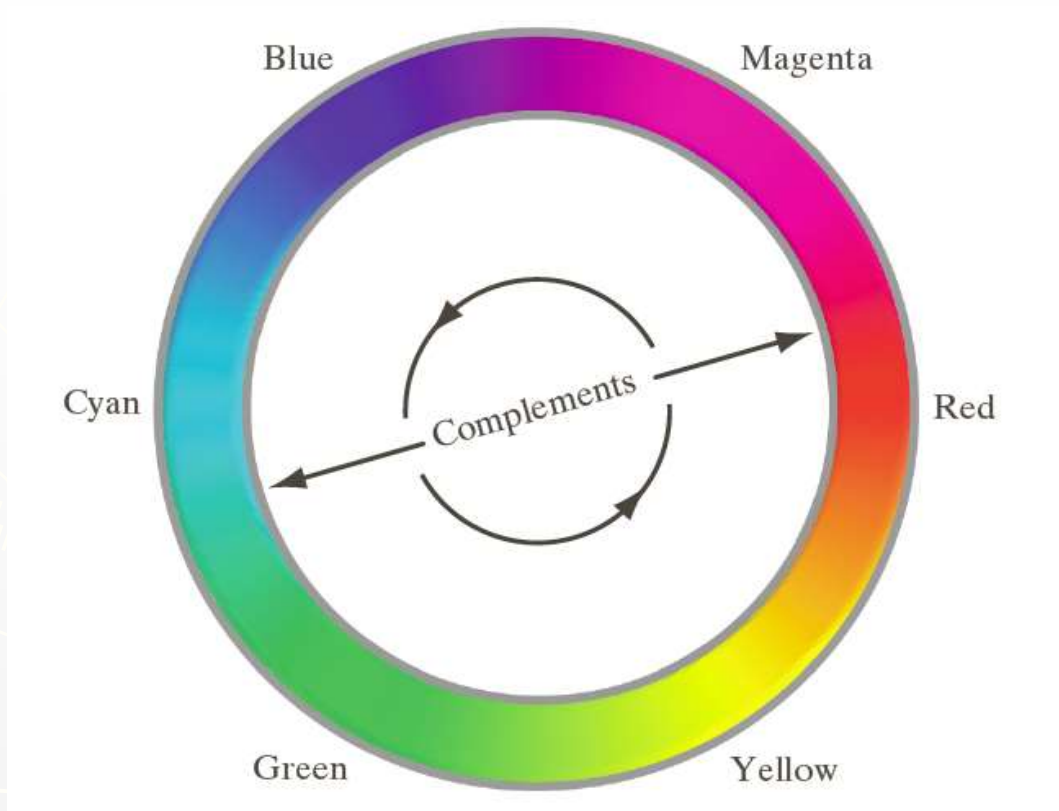
$$s_i = kr_i,$$

$$i = 1, 2, 3.$$

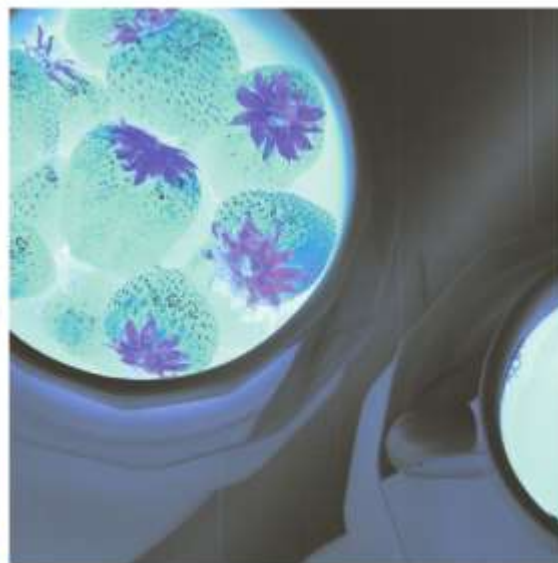
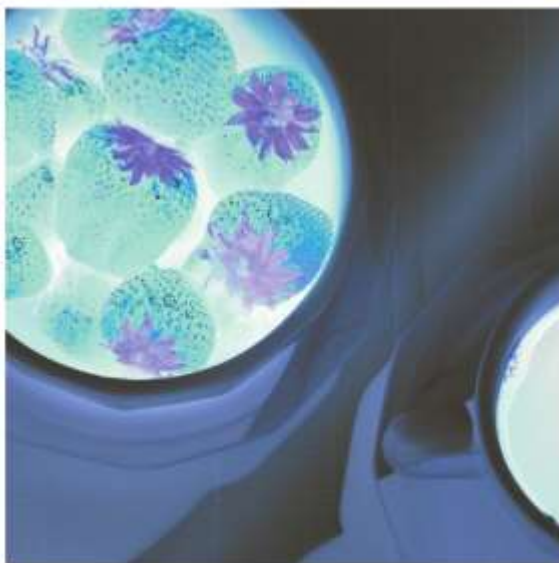
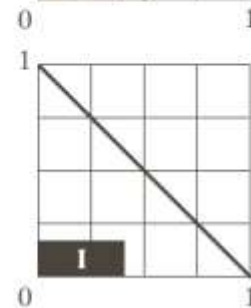
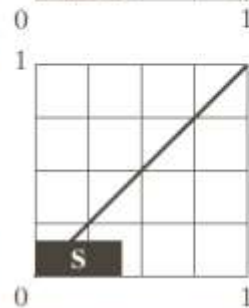
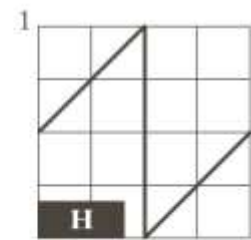
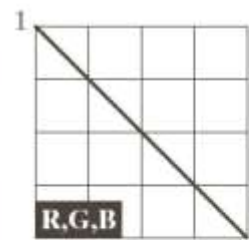
$$s_i = kr_i + (1 - k),$$

$$i = 1, 2, 3.$$

$$s_3 = kr_3$$



**FIGURE 6.32**  
Complements on  
the color circle.



# Color Slicing

- ▶ Highlighting a specific range of colors in an image

If the colors of interest are enclosed by a cube of width  $W$  and centered at a prototypical color with components  $(a_1, a_2, \dots, a_n)$ , the necessary set of transformations is

$$s_i = \begin{cases} 0.5 & \text{if } \left[ |r_j - a_j| > W / 2 \right]_{\text{any } 1 \leq j \leq n} \\ r_i & \text{otherwise} \end{cases}$$

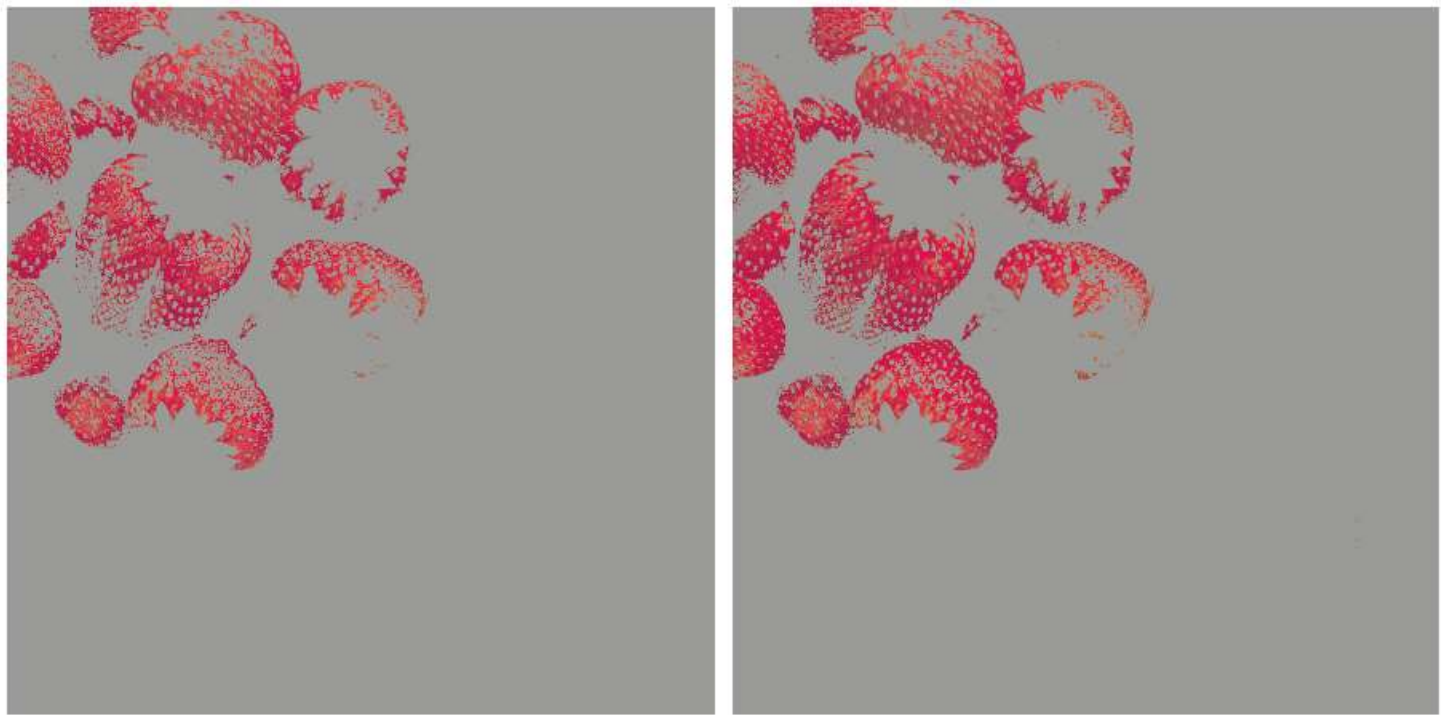
# Color Slicing

If a sphere is used to specify the colors of interest,  
 $R_0$  is the radius of the enclosing of its center.

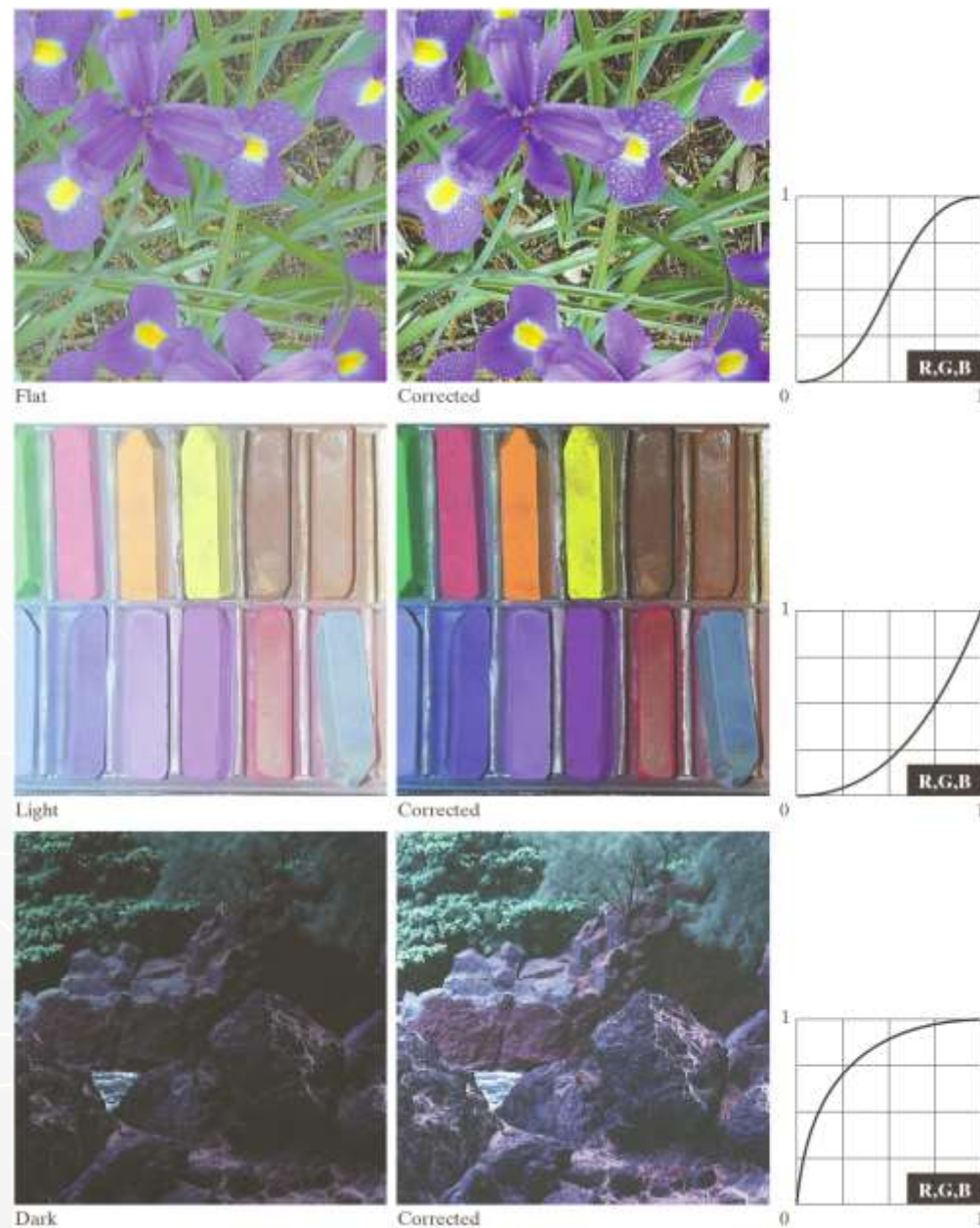
The transformations is

$$s_i = \begin{cases} 0.5 & \text{if } \sum_{j=1}^n (r_j - a_j)^2 > R_0^2 \\ r_i & \text{otherwise} \end{cases}$$

# Color Slicing



# Tone and Color Corrections

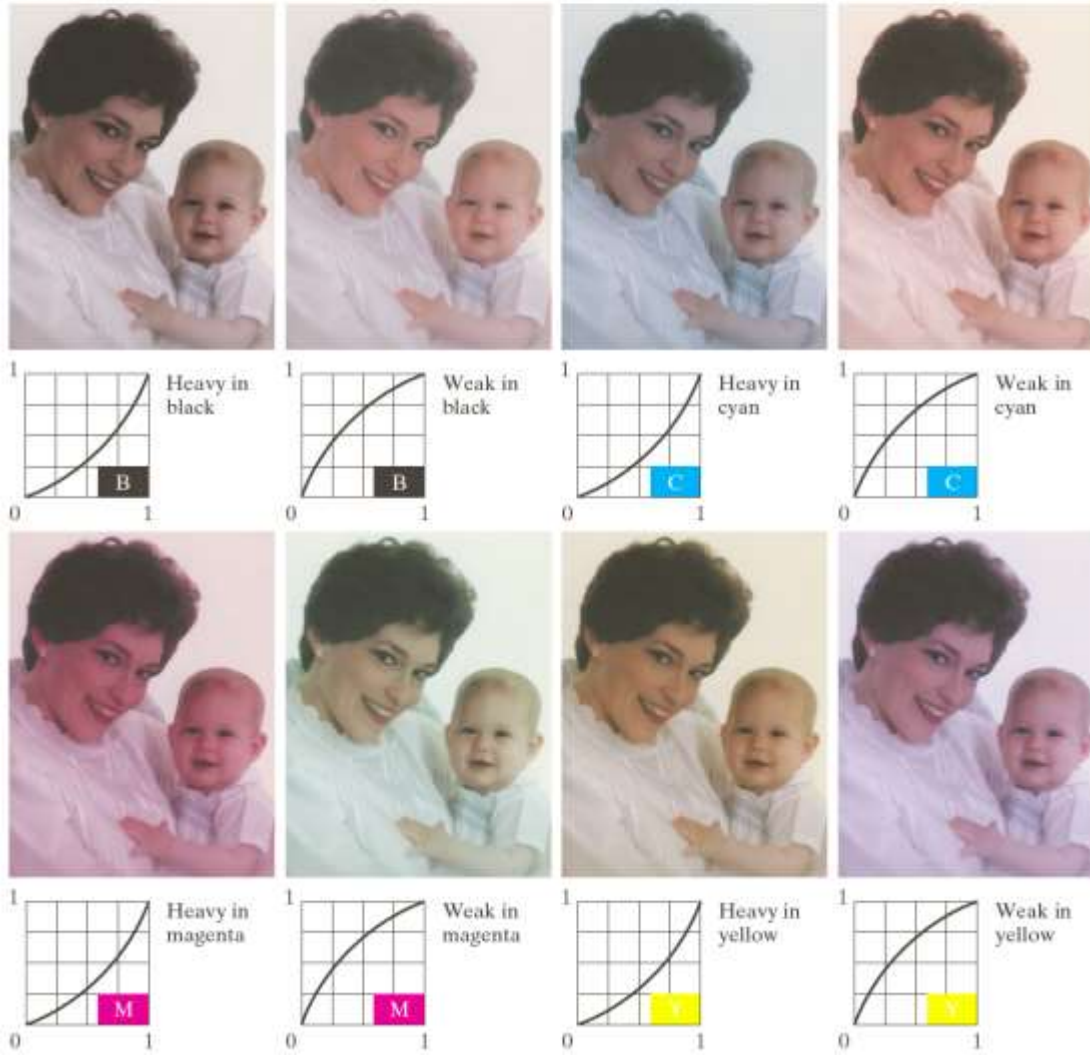


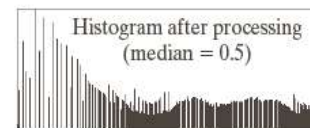
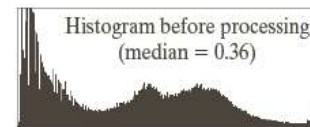
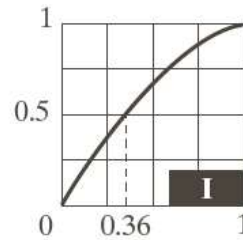
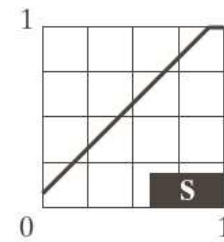
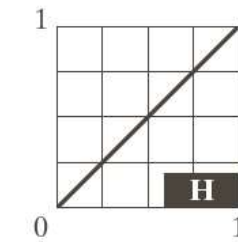
**FIGURE 6.35** Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.



Original/Corrected

**FIGURE 6.36** Color balancing corrections for CMYK color images.





|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 6.37**  
Histogram equalization (followed by saturation adjustment) in the HSI color space.



# Color Image Smoothing

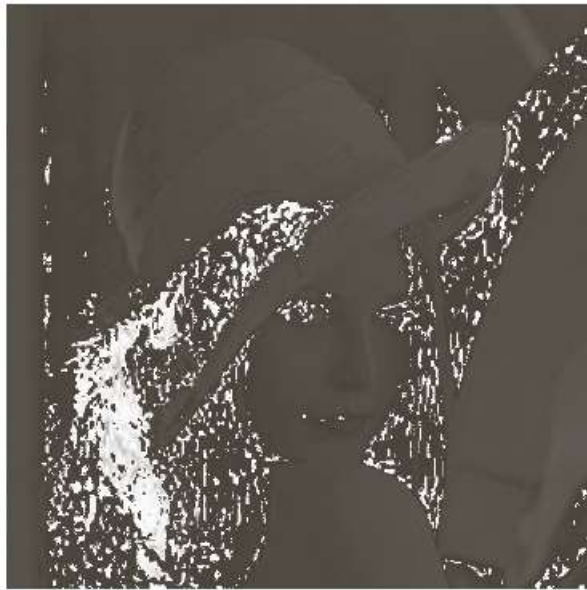
Let  $S_{xy}$  denote the set of coordinates defining a neighborhood centered at  $(x, y)$  in an RGB color image. The average of the RGB component vectors in this neighborhood is

$$\bar{c}(x, y) = \frac{1}{K} \sum_{(s,t) \in S_{xy}} c(s, t) = \begin{bmatrix} \frac{1}{K} \sum_{(s,t) \in S_{xy}} R(s, t) \\ \frac{1}{K} \sum_{(s,t) \in S_{xy}} G(s, t) \\ \frac{1}{K} \sum_{(s,t) \in S_{xy}} B(s, t) \end{bmatrix}$$



|   |   |
|---|---|
| a | b |
| c | d |

**FIGURE 6.38**  
(a) RGB image.  
(b) Red component image.  
(c) Green component. (d) Blue component.



a b c

**FIGURE 6.39** HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.



a b c

**FIGURE 6.40** Image smoothing with a  $5 \times 5$  averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

**Questions???**

**Thank You!!!**

